Mathematical Analysis I

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Author
MUHO CHOI
Email: sihyeon8240@gmail.com
GitHub: https://github.com/sihyeon8240

Source Code & Contributions
GitHub: https://github.com/sihyeon8240/math
Feel free to open issues or pull requests for corrections.

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

  $$(P \lor Q) \land R$$

  is evaluated by first computing $P \lor Q$.

- The **logical connectives** are interpreted in the order

  $$\neg \;>\; \land \;>\; \oplus \;>\; \lor \;>\; \implies, \;\impliedby \;>\; \iff$$

  Hence

  $$P \oplus Q \implies \neg R \lor S \land T \iff U \impliedby V$$

  is understood as

  $$((P \oplus Q) \implies ((\neg R) \lor (S \land T))) \iff (U \impliedby V)$$

- A **quantifier** binds the entire formula that follows it. Thus

  $$\forall x, P(x) \land Q(x) \implies R(x)$$

  means

  $$\forall x \, ((P(x) \land Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

  $$\begin{array}{c} P \\ \Downarrow \\ Q \end{array}$$

  merely expresses that P, and therefore Q in the current discussion; it is not itself a proposition.

Chapter 1

The Number Systems

1.1 The Natural Number System

Axiom 1.1.1. The set of **natural numbers** $\mathbb{N}$ satisfies the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- $\forall S \subseteq \mathbb{N}, (1 \in S \wedge (\forall n \in S, s(n) \in S)) \implies S = \mathbb{N}$

$s(n)$ is called the **successor** of n. These axioms are called the **Peano axioms**.

Remark. The **Peano axioms** do not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** are not discussed, so we will just assume that $\mathbb{N}$ exists.

Theorem 1.1.2. *Suppose P is a predicate with domain $\mathbb{N}$. Then*

$$P(1) \wedge (\forall n \in \mathbb{N}, P(n) \implies P(s(n))) \implies \forall n \in \mathbb{N}, P(n)$$

This method of proof is called the ***principle of mathematical induction****.*

Proof. Suppose

- $P(1)$

- $\forall n \in \mathbb{N}, P(n) \implies P(s(n))$

Define

$$S \coloneqq \{n \in \mathbb{N} \mid P(n)\} \subseteq \mathbb{N}$$

Then

- $1 \in S$
- $\forall n \in S, s(n) \in S$

By axiom 1.1.1,

$$\begin{gathered} S = \mathbb{N} \\ \Downarrow \\ \forall n \in \mathbb{N}, P(n) \end{gathered}$$

□

Definition 1.1.3. Suppose $n, m \in \mathbb{N}$.

- **Addition** $+$ on $\mathbb{N}$ is defined by
  - $n + 1 \coloneqq s(n)$
  - $n + s(m) \coloneqq s(n + m)$
- **Multiplication** $\cdot$ on $\mathbb{N}$ is defined by
  - $n \cdot 1 \coloneqq n$
  - $n \cdot s(m) \coloneqq n \cdot m + n$

Remark. Strictly speaking, definition 1.1.3 requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operations on $\mathbb{N}$ are **well-defined**.

Theorem 1.1.4. $\mathbb{N}$ *has the* ***well-ordering property****, which states*

$$\forall S \subseteq \mathbb{N}, S \neq \varnothing \implies \exists m \in S, m = \min S$$

Proof. Assume

$$\exists S \subseteq \mathbb{N}, S \neq \varnothing \wedge (\nexists m \in S, m = \min S)$$

Define

- $T \coloneqq \mathbb{N} \setminus S \subseteq \mathbb{N}$

- $L := \{n \in \mathbb{N} \mid \{1, 2, \ldots, n\} \subseteq T\} \subseteq \mathbb{N}$

We will show

$$\begin{gathered} L = \mathbb{N} \\ \Downarrow \\ T = \mathbb{N} \\ \Downarrow \\ S = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

- **Base case:** Assume $1 \in S$. Then

$$\begin{gathered} 1 = \min S \\ \Downarrow \\ \perp \end{gathered}$$

  Therefore,

$$\begin{gathered} 1 \notin S \\ \Downarrow \\ 1 \in T \\ \Downarrow \\ 1 \in L \end{gathered}$$

- **Inductive step:** Suppose $n \in L$. Then

$$\begin{gathered} n \in L \\ \Downarrow \\ \{1, 2, \ldots, n\} \subseteq T \\ \Downarrow \\ \{1, 2, \ldots, n\} \cap S = \varnothing \end{gathered}$$

  Assume $n + 1 \in S$. Then

$$\begin{gathered} n + 1 = \min S \\ \Downarrow \\ \perp \end{gathered}$$

  Therefore,

$$\begin{gathered} n + 1 \notin S \\ \Downarrow \\ n + 1 \in T \\ \Downarrow \\ \{1, 2, \ldots, n + 1\} \subseteq T \\ \Downarrow \\ n + 1 \in L \end{gathered}$$

By theorem 1.1.2,

$$L = \mathbb{N}$$

□

1.2 Relations

Definition 1.2.1. A **relation** R from A to B is defined by

$$R \subseteq A \times B$$

In particular, a relation R on A is defined by

$$R \subseteq A \times A$$

The notation xRy denotes $(x, y) \in R$.

Definition 1.2.2. A relation $\sim$ on S is called an **equivalence relation** if it satisfies

- **Reflexivity**: $\forall x \in S, x \sim x$
- **Symmetry**: $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity**: $\forall x, y, z \in S, x \sim y \land y \sim z \implies x \sim z$

Definition 1.2.3. A collection $\{A_\alpha\}_{\alpha \in I}$ is called a **partition** of S if

- **Nonemptiness**: $\forall \alpha \in I, \varnothing \neq A_\alpha \subseteq S$
- **Disjointness**: $\forall \alpha, \beta \in I, \alpha \neq \beta \implies A_\alpha \cap A_\beta = \varnothing$
- **Exhaustivity**: $\displaystyle\bigcup_{\alpha \in I} A_\alpha = S$

Definition 1.2.4. Suppose

- $\sim$ is an equivalence relation on S
- $x \in S$

The **equivalence class** of x is defined by

$$[x] \coloneqq \{y \in S \mid y \sim x\}$$

The set of all equivalence classes of S is denoted by

$$S/\sim \coloneqq \{[x] \mid x \in S\}$$

Lemma 1.2.5. *Suppose $\sim$ is an equivalence relation on S. Then*

$$\forall x, y \in S, [x] \cap [y] \neq \varnothing \implies [x] = [y]$$

Proof. Suppose

- $x, y \in S$
- $[x] \cap [y] \neq \varnothing$

Then

$$\begin{gathered} \exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y \end{gathered}$$

Therefore,

$$\begin{gathered} \forall w \in S, w \sim x \implies w \sim y \\ \Downarrow \\ \forall w \in [x], w \in [y] \\ \Downarrow \\ [x] \subseteq [y] \end{gathered}$$

Similarly,

$$[y] \subseteq [x]$$

□

Theorem 1.2.6. *Suppose $\sim$ is an equivalence relation on S. Then*

$$\{[\alpha]\}_{[\alpha] \in S/\sim} = S/\sim$$

is a partition of S.

Proof. By definition 1.2.3,

- **Nonemptiness**:
$$\forall[\alpha] \in S/\sim, \alpha \in [\alpha] \subseteq S$$

- **Disjointness**: By lemma 1.2.5,
$$\forall[\alpha],[\beta] \in S/\sim, [\alpha] \neq [\beta] \implies [\alpha] \cap [\beta] = \varnothing$$

- **Exhaustivity**:
  - ($\subseteq$)
$$\begin{gathered}\forall[\alpha] \in S/\sim, [\alpha] \subseteq S \\ \Downarrow \\ \bigcup_{[\alpha]\in S/\sim} [\alpha] \subseteq S\end{gathered}$$
  - ($\supseteq$)
$$\begin{gathered}\forall\beta \in S, [\beta] \in S/\sim \\ \Downarrow \\ \forall\beta \in S, \beta \in [\beta] \subseteq \bigcup_{[\alpha]\in S/\sim} [\alpha]\end{gathered}$$

□

Theorem 1.2.7. *Suppose $\{A_\alpha\}_{\alpha\in I}$ is a partition of S. Define $\sim$ on S by*
$$\forall x, y \in S, x \sim y \iff \exists\alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$$
Then

(A) *$\sim$ is an equivalence relation on S*

(B)
- $\forall x \in S, \exists\beta \in I, [x] = A_\beta$
- $\forall\beta \in I, \exists x \in S, [x] = A_\beta$

(C) $\{A_\alpha\}_{\alpha\in I} = S/\sim$

Proof.

(A) By definition 1.2.2,

- **Reflexivity**: Suppose $x \in S$. Then

$$\begin{gathered} x \in \bigcup_{\alpha \in I} A_\alpha \\ \Downarrow \\ \exists \beta \in I, x \in A_\beta \\ \Downarrow \\ x \sim x \end{gathered}$$

- **Symmetry**: Suppose
  - $x, y \in S$
  - $x \sim y$

  Then

$$\begin{gathered} \exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha \\ \Downarrow \\ y \in A_\alpha \wedge x \in A_\alpha \\ \Downarrow \\ y \sim x \end{gathered}$$

- **Transitivity**: Suppose
  - $x, y, z \in S$
  - $x \sim y$
  - $y \sim z$

  Then
  - $\exists \alpha \in I, x \in A_\alpha \wedge y \in A_\alpha$
  - $\exists \beta \in I, y \in A_\beta \wedge z \in A_\beta$

  Therefore,

$$\begin{gathered} y \in A_\alpha \wedge y \in A_\beta \\ \Downarrow \\ A_\alpha \cap A_\beta \neq \varnothing \\ \Downarrow \\ \alpha = \beta \\ \Downarrow \\ x \in A_\alpha \wedge z \in A_\alpha \\ \Downarrow \\ x \sim z \end{gathered}$$

(B)

- Suppose $x \in S$. Then

$$\begin{gathered} x \in \bigcup_{\alpha \in I} A_\alpha \\ \Downarrow \\ \exists \beta \in I, x \in A_\beta \end{gathered}$$

- Suppose $\beta \in I$. Then

$$\exists x \in A_\beta \subseteq S$$

We will show $[x] = A_\beta$.

- ($\subseteq$) Suppose $y \in [x]$. Then

$$\begin{gathered}\exists \gamma \in I, y \in A_\gamma \wedge x \in A_\gamma \\ \Downarrow \\ x \in A_\gamma \cap A_\beta \\ \Downarrow \\ \gamma = \beta \\ \Downarrow \\ y \in A_\beta\end{gathered}$$

- ($\supseteq$) Suppose $y \in A_\beta$. Then

$$\begin{gathered}y \sim x \\ \Downarrow \\ y \in [x]\end{gathered}$$

(C) By theorem 1.2.7,

- ($\subseteq$)
$$\forall A_\beta \in \{A_\alpha\}_{\alpha \in I}, \exists x \in S, A_\beta = [x] \in S/\sim$$

- ($\supseteq$)
$$\forall [x] \in S/\sim, \exists \beta \in I, [x] = A_\beta \in \{A_\alpha\}_{\alpha \in I}$$

□

1.3 The Integer and Rational Number Systems

Theorem 1.3.1. *Define $\sim$ on $\mathbb{N} \times \mathbb{N}$ by*

$$\forall (a, b), (c, d) \in \mathbb{N} \times \mathbb{N}, (a, b) \sim (c, d) \iff a + d = b + c$$

Then $\sim$ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Proof. By definition 1.2.2,

- **Reflexivity**: Suppose $(a, b) \in \mathbb{N} \times \mathbb{N}$. Then
$$\begin{gathered} a + b = b + a \\ \Downarrow \\ (a, b) \sim (a, b) \end{gathered}$$
- **Symmetry**: Suppose
  - $(a, b), (c, d) \in \mathbb{N} \times \mathbb{N}$
  - $(a, b) \sim (c, d)$

  Then
$$\begin{gathered} a + d = b + c \\ \Downarrow \\ c + b = d + a \\ \Downarrow \\ (c, d) \sim (a, b) \end{gathered}$$
- **Transitivity**: Suppose
  - $(a, b), (c, d), (e, f) \in \mathbb{N} \times \mathbb{N}$
  - $(a, b) \sim (c, d)$
  - $(c, d) \sim (e, f)$

  Then
$$\begin{gathered} a + d = b + c \wedge c + f = d + e \\ \Downarrow \\ a + d + f = b + c + f = b + d + e \\ \Downarrow \\ a + f = b + e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

□

Remark. Strictly speaking, theorem 1.3.1 requires several **properties** of operations on $\mathbb{N}$. In this textbook, the properties are not discussed, so just **accepted** as true. The same issue arises for theorem 1.3.4.

Definition 1.3.2. Recall the relation $\sim$ defined in theorem 1.3.1. The set of **integers** $\mathbb{Z}$ is defined by
$$\mathbb{Z} := (\mathbb{N} \times \mathbb{N}) / \sim$$

Definition 1.3.3. Suppose $[(a, b)], [(c, d)] \in \mathbb{Z}$.

- **Addition** $+$ on $\mathbb{Z}$ is defined by

$$[(a, b)] + [(c, d)] \coloneqq [(a + c, b + d)]$$

- **Multiplication** $\cdot$ on $\mathbb{Z}$ is defined by

$$[(a, b)] \cdot [(c, d)] \coloneqq [(a \cdot c + b \cdot d, a \cdot d + b \cdot c)]$$

Remark. The operations on $\mathbb{Z}$ are **well-defined**; that is, they do not depend on the **choice of representatives** of the equivalence classes. Indeed, if

- $(a, b) \sim (a', b')$
- $(c, d) \sim (c', d')$

then one can verify that

- $(a + c, b + d) \sim (a' + c', b' + d')$
- $(a \cdot c + b \cdot d, a \cdot d + b \cdot c) \sim (a' \cdot c' + b' \cdot d', a' \cdot d' + b' \cdot c')$

The same issue arises for operations on $\mathbb{Q}$, whose **well-definedness** could be verified in exactly the same manner.

Theorem 1.3.4. *Define*

$$\mathbb{Z}^+ \coloneqq \{[(n + 1, 1)] \in \mathbb{Z} \mid n \in \mathbb{N}\} \subseteq \mathbb{Z}$$

Define $\sim$ *on* $\mathbb{Z} \times \mathbb{Z}^+$ *by*

$$\forall (a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}^+, (a, b) \sim (c, d) \iff a \cdot d = b \cdot c$$

Then $\sim$ *is an equivalence relation on* $\mathbb{Z} \times \mathbb{Z}^+$.

Proof. By definition 1.2.2,

- **Reflexivity**: Suppose $(a, b) \in \mathbb{Z} \times \mathbb{Z}^+$. Then

$$\begin{gathered} a \cdot b = b \cdot a \\ \Downarrow \\ (a, b) \sim (a, b) \end{gathered}$$

- **Symmetry**: Suppose
  - $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}^+$
  - $(a, b) \sim (c, d)$

  Then
$$\begin{gathered} a \cdot d = b \cdot c \\ \Downarrow \\ c \cdot b = d \cdot a \\ \Downarrow \\ (c, d) \sim (a, b) \end{gathered}$$

- **Transitivity**: Suppose
  - $(a, b), (c, d), (e, f) \in \mathbb{Z} \times \mathbb{Z}^+$
  - $(a, b) \sim (c, d)$
  - $(c, d) \sim (e, f)$

  Then
$$\begin{gathered} a \cdot d = b \cdot c \wedge c \cdot f = d \cdot e \\ \Downarrow \\ a \cdot d \cdot f = b \cdot c \cdot f = b \cdot d \cdot e \\ \Downarrow \\ a \cdot f = b \cdot e \\ \Downarrow \\ (a, b) \sim (e, f) \end{gathered}$$

□

Definition 1.3.5. Recall the relation $\sim$ defined in theorem 1.3.4. The set of **rational numbers** $\mathbb{Q}$ is defined by
$$\mathbb{Q} := (\mathbb{Z} \times \mathbb{Z}^+)/\sim$$

Definition 1.3.6. Suppose $[(a, b)], [(c, d)] \in \mathbb{Q}$.

- **Addition** $+$ on $\mathbb{Q}$ is defined by
$$[(a, b)] + [(c, d)] := [(ad + bc, bd)]$$

- **Multiplication** $\cdot$ on $\mathbb{Q}$ is defined by
$$[(a, b)] \cdot [(c, d)] := [(ac, bd)]$$

Remark. Strictly speaking, the constructions of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ produce **different kind** of sets. For example,

- $n \in \mathbb{N}$
- $[(n+1,1)] \in \mathbb{Z}$
- $[(n,[(2,1)])] \in \mathbb{Q}$

are distinct objects in the underlying set-theoretic construction. Nevertheless, there are **canonical embeddings**

- $\mathbb{N} \to \mathbb{Z}$ such that $n \mapsto [(n+1,1)]$
- $\mathbb{Z} \to \mathbb{Q}$ such that $a \mapsto [(a,[(2,1)])]$

which preserve the **algebraic operations**. Therefore, we identify each number system with its image in the next larger one and write

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$$

This identification is justified by the fact that the embeddings are canonical, requiring **no arbitrary choices**. Throughout this textbook, inclusions such as

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

should be understood as canonical identifications rather than literal inclusions arising directly from the set-theoretic constructions.

Remark. Define

$$0 \coloneqq [(1,1)] \in \mathbb{Z}$$

The notations below denote subsets of $\mathbb{Z}$:

- $[0,\infty)_{\mathbb{N}} \coloneqq \mathbb{N} \cup \{0\} \subseteq \mathbb{Z}$
- $\forall n \in [0,\infty)_{\mathbb{N}}, [n,\infty)_{\mathbb{N}} \coloneqq \{k \in [0,\infty)_{\mathbb{N}} \mid k \geq n\}$
- $\forall n,m \in [0,\infty)_{\mathbb{N}}, [n,m]_{\mathbb{N}} \coloneqq \{k \in [n,\infty)_{\mathbb{N}} \mid k \leq m\}$

1.4 Ordered Sets

Definition 1.4.1. Suppose a relation $<$ on S satisfies the following axioms:

- $\forall x, y, z \in S, x < y \wedge y < z \implies x < z$
- $\forall x, y \in S, x \neq y \implies x < y \oplus y < x$

Then

- The relation $<$ is called an **(total) order** on S
- The structure $(S, <)$ is called an **(totally) ordered set**

The notation

- $x > y$ denotes $y < x$
- $x \geq y$ denotes $x > y \vee x = y$
- $x \leq y$ denotes $y \geq x$

Remark. An ordered set $(S, <)$ is a **structure**, distinct from its **underlying set** S. For any structure $\mathcal{S}$, the notation $|\mathcal{S}|$ denotes its underlying set, not its **cardinality**. That is,

$$S = |(S, <)| \neq (S, <)$$

Example 1.4.2. $(\mathbb{N}, <), (\mathbb{Z}, <), (\mathbb{Q}, <)$ are ordered sets with

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N}, n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+, n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+, p = q + r$

where

$$\mathbb{Q}^+ := \left\{ [(p, q)] \in \mathbb{Q} \;\middle|\; p, q \in \mathbb{Z}^+ \right\} \subseteq \mathbb{Q}$$

Throughout this textbook, the **number systems** use the **standard order relations** above, unless otherwise specified.

Definition 1.4.3. Suppose $E \subseteq |(S, <)|$. Then

- An element $m \in S$ is called a **lower bound** of E if
$$\forall x \in E, m \leq x$$
The set of all lower bounds of E is denoted by
$$E^\ell := \{m \in S \mid \forall x \in E, m \leq x\}$$
E is said to be **bounded below** if
$$E^\ell \neq \varnothing$$
- An element $M \in S$ is called an **upper bound** of E if
$$\forall x \in E, x \leq M$$
The set of all upper bounds of E is denoted by
$$E^u := \{M \in S \mid \forall x \in E, x \leq M\}$$
E is said to be **bounded above** if
$$E^u \neq \varnothing$$

Definition 1.4.4. Suppose $E \subseteq |(S, <)|$. Then

- An element $m \in E$ is called a **minimum** of E if
$$m \in E^\ell$$
Such an m is denoted by
$$m = \min E$$
- An element $M \in E$ is called a **maximum** of E if
$$M \in E^u$$
Such an M is denoted by
$$M = \max E$$

Theorem 1.4.5.

(A) $\forall E \subseteq \mathbb{Z}, E \neq \varnothing \land E^\ell \neq \varnothing \implies \exists m \in E, m = \min E$

(B) $\forall E \subseteq \mathbb{Z}, E \neq \varnothing \land E^u \neq \varnothing \implies \exists M \in E, M = \max E$

Proof. Suppose

- $E \subseteq \mathbb{Z}$
- $E \neq \varnothing$

Then

(A) Suppose $E^\ell \neq \varnothing$. Then

$$\exists k \in \mathbb{Z}, \forall \alpha \in E, k \leq \alpha$$

Define

$$F := \{\alpha - k + 1 \mid \alpha \in E\} \subseteq \mathbb{N}$$

By theorem 1.1.4,

$$\begin{gathered} \exists n \in F, n = \min F \\ \Downarrow \\ \forall \alpha \in E, n \leq \alpha - k + 1 \\ \Downarrow \\ \forall \alpha \in E, n + k - 1 \leq \alpha \end{gathered}$$

Therefore,

$$\begin{gathered} \exists m \in E, n = m - k + 1 \\ \Downarrow \\ m = n + k - 1 = \min E \end{gathered}$$

(B) Suppose $E^u \neq \varnothing$. Then

$$\exists k \in \mathbb{Z}, \forall \alpha \in E, \alpha \leq k$$

Define

$$F := \{-\alpha + k + 1 \mid \alpha \in E\} \subseteq \mathbb{N}$$

By theorem 1.1.4,

$$\begin{gathered} \exists n \in F, n = \min F \\ \Downarrow \\ \forall \alpha \in E, n \leq -\alpha + k + 1 \\ \Downarrow \\ \forall \alpha \in E, \alpha \leq k - n + 1 \end{gathered}$$

Therefore,

$$\begin{gathered} \exists M \in E, n = -M + k + 1 \\ \Downarrow \\ M = k - n + 1 = \max E \end{gathered}$$

$\square$

Definition 1.4.6. Suppose $E \subseteq |(S, <)|$. Then

- Suppose $E^u \neq \varnothing$. An element $\alpha \in S$ is called a **supremum** of E if
$$\alpha = \min E^u$$
Such an α is denoted by
$$\alpha = \sup E$$
- Suppose $E^\ell \neq \varnothing$. An element $\alpha \in S$ is called an **infimum** of E if
$$\alpha = \max E^\ell$$
Such an α is denoted by
$$\alpha = \inf E$$

Definition 1.4.7. An ordered set $(S, <)$ is said to have the **least upper bound property** if
$$\forall E \subseteq S, E \neq \varnothing \wedge E^u \neq \varnothing \implies \exists \alpha \in S, \alpha = \sup E$$

Example 1.4.8. Define

- $S := \left\{p \in \mathbb{Q}^+ \;\middle|\; p^2 < 2\right\} \subseteq \mathbb{Q}$
- $\forall p \in \mathbb{Q}^+, q := p - \dfrac{p^2 - 2}{p + 2} = \dfrac{2p + 2}{p + 2}$

Then

- If $p \in S$, then
$$\begin{aligned} q^2 - 2 = \frac{2(p^2 - 2)}{(p+2)^2} &< 0 \\ &\Downarrow \\ q &\in S \\ &\Downarrow && q > p \\ p &\notin S^u \end{aligned}$$
- Otherwise,
$$\begin{aligned} p^2 &\geq 2 \\ &\Downarrow && \nexists p \in \mathbb{Q}^+, p^2 = 2 \\ p^2 &> 2 \end{aligned}$$
Then
$$\begin{aligned} q^2 - 2 = \frac{2(p^2 - 2)}{(p+2)^2} &> 0 \\ &\Downarrow \\ q &\in S^u \\ &\Downarrow && q < p \\ p &\neq \min S^u \end{aligned}$$

Therefore, $(\mathbb{Q}, <)$ does not have the least upper bound property.

Theorem 1.4.9. *Suppose* $(S, <)$ *have the least upper bound property. Then*

$$\forall E \subseteq S, E \neq \varnothing \wedge E^{\ell} \neq \varnothing \implies \exists \alpha \in S, \alpha = \inf E$$

Proof. Suppose

- $E \subseteq S$
- $E \neq \varnothing$
- $E^{\ell} \neq \varnothing$

Then

$$\begin{gathered} \exists x \in E \\ \Downarrow \\ \forall y \in E^{\ell}, y \leq x \\ \Downarrow \\ x \in \left(E^{\ell}\right)^{u} \\ \Downarrow \\ \exists \alpha \in S, \alpha = \sup E^{\ell} \end{gathered}$$

Assume $\alpha \notin E^{\ell}$. Then

$$\begin{gathered} \exists \beta \in E, \beta < \alpha \\ \Downarrow \qquad \alpha = \sup E^{\ell} \\ \beta \notin \left(E^{\ell}\right)^{u} \\ \Downarrow \\ \exists \gamma \in E^{\ell}, \beta < \gamma \\ \Downarrow \qquad \beta \in E \\ \gamma \leq \beta \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\begin{gathered} \alpha \in E^{\ell} \\ \Downarrow \qquad \alpha = \sup E^{\ell} \\ \alpha = \max E^{\ell} \end{gathered}$$

$\square$

1.5 Fields and Ordered Fields

Definition 1.5.1. Suppose two operations

- **Addition** $+$
- **Multiplication** $\cdot$

on F satisfy the following axioms:

(A) Axioms for **addition**

- $\forall x, y \in F, x + y \in F$
- $\forall x, y \in F, x + y = y + x$
- $\forall x, y, z \in F, (x + y) + z = x + (y + z)$
- $\exists \alpha \in F, \forall x \in F, x + \alpha = x$
- $\forall x \in F, \exists \beta \in F, x + \beta = \alpha$

(B) Axioms for **multiplication**

- $\forall x, y \in F, x \cdot y \in F$
- $\forall x, y \in F, x \cdot y = y \cdot x$
- $\forall x, y, z \in F, (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- $\exists \gamma \in F, \gamma \neq \alpha \wedge (\forall x \in F, x \cdot \gamma = x)$
- $\forall x \in F, x \neq \alpha \implies \exists \omega \in F, x \cdot \omega = \gamma$

(C) The **distributive law**

- $\forall x, y, z \in F, x \cdot (y + z) = x \cdot y + x \cdot z$

Then the structure $(F, +, \cdot)$ is called a **field**.

Definition 1.5.2. Suppose

- $(F, +, \cdot)$
- $(F, <)$

satisfy the following axioms:

- $\forall x, y, z \in F, x < y \implies x + z < y + z$
- $\forall x, y \in F, x > 0 \wedge y > 0 \implies x \cdot y > 0$

Then the structure $(F, +, \cdot, <)$ is called an **ordered field**.

Remark. $(\mathbb{Q}, +, \cdot, <)$ is an ordered field. Also, there exists an ordered field $(\mathbb{R}, +, \cdot, <)$ which has the *least upper bound property*. In this textbook, the construction of $\mathbb{R}$ is not discussed, so we will just assume that $(\mathbb{R}, +, \cdot, <)$ exists and satisfies axiom 1.6.1.

1.6 The Real Number System

Axiom 1.6.1. $(\mathbb{R}, +, \cdot, <)$ has the *least upper bound property*.

Theorem 1.6.2. $(\mathbb{R}, +, \cdot, <)$ *has the* ***Archimedean property****, which states*

$$\forall x \in \mathbb{R}^+, \forall y \in \mathbb{R}, \exists n \in \mathbb{N}, nx > y$$

where

$$\mathbb{R}^+ := \{x \in \mathbb{R} \mid x > 0\}$$

Proof. Suppose

- $x \in \mathbb{R}^+$
- $y \in \mathbb{R}$

Define

$$E := \{nx \mid nx \leq y,\ n \in \mathbb{N}\} \subseteq \mathbb{R}$$

Then

- If $E = \varnothing$, then

$$\begin{gathered} 1 \cdot x \notin E \\ \Downarrow \\ 1 \cdot x > y \end{gathered}$$

- Otherwise,

$$\begin{gathered} E \neq \varnothing \wedge y \in E^u \\ \Downarrow \qquad \text{Axiom 1.6.1} \\ \exists \alpha \in \mathbb{R}, \alpha = \sup E \\ \Downarrow \\ \alpha - x \notin E^u \\ \Downarrow \\ \exists m \in \mathbb{N}, mx > \alpha - x \\ \Downarrow \\ (m+1)x > \alpha \\ \Downarrow \qquad \alpha = \sup E \\ (m+1)x \notin E \\ \Downarrow \\ (m+1)x > y \end{gathered}$$

$\square$

Lemma 1.6.3.

$$\forall x \in [-1, +\infty), \forall n \in \mathbb{N}, (1+x)^n \geq 1 + nx$$

*This inequality is called **Bernoulli's inequality**.*

Proof. Suppose $x \in [-1, +\infty)$. Define

$$\forall n \in \mathbb{N}, P(n) \iff (1+x)^n \geq 1 + nx$$

Then

- **Base Case:**
$$\begin{gathered}(1+x)^1 = 1 + 1 \cdot x \\ \Downarrow \\ P(1)\end{gathered}$$

- **Inductive step:** Suppose $P(n)$. Then
$$\begin{gathered}(1+x)^n \geq 1 + nx \\ \Downarrow \\ (1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \\ \Downarrow \\ (1+x)^{n+1} \geq 1 + (n+1)x \\ \Downarrow \\ P(n+1)\end{gathered} \qquad \begin{gathered} 1+x \geq 0 \\ \\ x^2 \geq 0 \end{gathered}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n)$$

□

Theorem 1.6.4.

$$\forall \varepsilon \in \mathbb{R}^+, \forall b \in (1, +\infty), \exists n \in \mathbb{N}, b^{-n} < \varepsilon$$

Proof. Suppose

- $\varepsilon \in \mathbb{R}^+$
- $b \in (1, +\infty)$

Then

$$\begin{array}{c} b-1\in\mathbb{R}^+ \\ \Downarrow \\ \exists n\in\mathbb{N}, n(b-1)>\frac{1}{\varepsilon} \\ \Downarrow \\ (1+(b-1))^n\geq 1+n(b-1)>\frac{1}{\varepsilon} \\ \Downarrow \\ (1+(b-1))^{-n}<\varepsilon \\ \Downarrow \\ b^{-n}<\varepsilon \end{array}$$

Theorem 1.6.2

Lemma 1.6.3

□

Theorem 1.6.5.

(A) $\forall E\subseteq\mathbb{Z}, E\neq\varnothing\wedge(\exists\alpha\in\mathbb{R},\forall x\in E,\alpha\leq x) \implies \exists m\in E, m=\min E$

(B) $\forall E\subseteq\mathbb{Z}, E\neq\varnothing\wedge(\exists\beta\in\mathbb{R},\forall x\in E,x\leq\beta) \implies \exists M\in E, M=\max E$

Proof. Suppose

- $E\subseteq\mathbb{Z}$
- $E\neq\varnothing$

Then

(A) Suppose

$$\exists\alpha\in\mathbb{R},\forall x\in E,\alpha\leq x$$

Then

- If $\alpha\geq 0$, then

$$\begin{array}{c} \forall x\in E, 0\leq\alpha\leq x \\ \Downarrow \\ 0\in E^\ell \end{array}$$

- Otherwise, by theorem 1.6.2,

$$\begin{array}{c} \exists n\in\mathbb{N}, n>-\alpha \\ \Downarrow \\ \forall x\in E, -n<\alpha\leq x \\ \Downarrow \\ -n\in E^\ell \end{array}$$

Therefore,

$$\begin{array}{c} E^{\ell} \neq \varnothing \\ \Downarrow \\ \exists m \in E, m = \min E \end{array} \qquad \text{Theorem 1.4.5}$$

(B) Suppose

$$\exists \beta \in \mathbb{R}, \forall x \in E, x \leq \beta$$

By theorem 1.6.2,

$$\begin{array}{c} \exists n \in \mathbb{N}, n > \beta \\ \Downarrow \\ \forall x \in E, x \leq \beta < n \\ \Downarrow \\ n \in E^{u} \\ \Downarrow \\ \exists M \in E, M = \max E \end{array} \qquad \text{Theorem 1.4.5}$$

□

Definition 1.6.6. Suppose $x \in \mathbb{R}$.

- The **floor function** $\lfloor x \rfloor$ is defined by

$$\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$$

- The **ceiling function** $\lceil x \rceil$ is defined by

$$\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$$

By theorem 1.6.5,

$$\forall x \in \mathbb{R}, \exists k, m \in \mathbb{Z}, k = \lfloor x \rfloor \wedge m = \lceil x \rceil$$

Theorem 1.6.7. *$\mathbb{Q}$ is **dense** in $\mathbb{R}$, which states*

$$\forall x, y \in \mathbb{R}, x < y \implies \exists p \in \mathbb{Q}, x < p < y$$

Proof. Suppose

- $x, y \in \mathbb{R}$
- $x < y$

Define

$$\varepsilon := y - x > 0$$

By theorem 1.6.2,

$$\exists n \in \mathbb{N}, \frac{1}{n} < \varepsilon$$

Define

$$S := \left\{ k \in \mathbb{Z} \,\middle|\, \frac{k}{n} \leq x \right\} \subseteq \mathbb{Z}$$

Then

$$\begin{gathered} \lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \\ \Downarrow \\ n\lfloor x \rfloor \leq nx < n\lfloor x \rfloor + n \\ \Downarrow \\ n\lfloor x \rfloor \in S \end{gathered}$$

Also,

$$\forall k \in S, k \leq nx$$

By theorem 1.6.5,

$$\exists M \in \mathbb{Z}, M = \max S$$

Then

$$\begin{gathered} M \in S \wedge M + 1 \notin S \\ \Downarrow \\ \frac{M}{n} \leq x < \frac{M+1}{n} \\ \Downarrow \\ x < \frac{M+1}{n} \leq x + \frac{1}{n} < x + \varepsilon = y \end{gathered}$$

□

Lemma 1.6.8.

$$\forall n \in \mathbb{N}, \forall a, b \in \mathbb{R}^+, a < b \implies b^n - a^n \leq (b-a)nb^{n-1}$$

Proof. Suppose

- $n \in \mathbb{N}$
- $a, b \in \mathbb{R}^+$
- $a < b$

Then

$$b^n - a^n = (b-a)\sum_{k=0}^{n-1} b^{n-1-k}a^k \leq (b-a)nb^{n-1}$$

□

Lemma 1.6.9.

$$\forall n \in \mathbb{N}, \forall a, b \in \mathbb{R}^+, a < b \iff a^n < b^n$$

Proof. Suppose

- $n \in \mathbb{N}$
- $a, b \in \mathbb{R}^+$

Then

$$b^n - a^n = (b-a)\sum_{k=0}^{n-1} b^{n-1-k}a^k$$
$$\Downarrow$$
$$a < b \iff a^n < b^n$$

□

Theorem 1.6.10.

$$\forall x \in \mathbb{R}^+, \forall n \in \mathbb{N}, \exists! y \in \mathbb{R}^+, y^n = x$$

Proof. Suppose

- $x \in \mathbb{R}^+$
- $n \in \mathbb{N}$

Define

$$E \coloneqq \left\{ t \in \mathbb{R}^+ \;\middle|\; t^n < x \right\} \subseteq \mathbb{R}^+$$

Then

- If $x \geq 1$, then

$$2^{-n} < 1 \leq x$$
$$\Downarrow$$
$$2^{-1} \in E$$

  Also,

$$\forall t \in E, t^n < x \leq x^n$$
$$\Downarrow \qquad \text{Lemma 1.6.9}$$
$$\forall t \in E, t < x$$
$$\Downarrow$$
$$x \in E^u$$

- Otherwise,

$$x < 1$$
$$\Downarrow$$
$$x^{2n} < x$$
$$\Downarrow$$
$$x^2 \in E$$

  Also,

$$\forall t \in E, t^n < x < 1 = 1^n$$
$$\Downarrow \qquad \text{Lemma 1.6.9}$$
$$\forall t \in E, t < 1$$
$$\Downarrow$$
$$1 \in E^u$$

Therefore,

$$E \neq \varnothing \wedge E^u \neq \varnothing$$
$$\Downarrow \qquad \text{Axiom 1.6.1}$$
$$\exists \alpha \in \mathbb{R}^+, \alpha = \sup E$$

We will show

$$\alpha^n \nless x \wedge \alpha^n \ngtr x$$
$$\Downarrow$$
$$\alpha^n = x$$

- Assume $\alpha^n < x$. By theorem 1.6.7,

$$\exists h \in \mathbb{Q}, 0 < h < \min\left\{1, \frac{x - \alpha^n}{n(\alpha+1)^{n-1}}\right\}$$
$$\Downarrow \qquad \text{Lemma 1.6.8}$$
$$(\alpha+h)^n - \alpha^n \leq hn(\alpha+h)^{n-1} < hn(\alpha+1)^{n-1}$$
$$\Downarrow$$
$$(\alpha+h)^n < \alpha^n + hn(\alpha+1)^{n-1} < \alpha^n + (x - \alpha^n) = x$$
$$\Downarrow$$
$$\alpha + h \in E$$
$$\Downarrow \qquad \alpha = \sup E$$
$$\perp$$

- Assume $\alpha^n > x$. By theorem 1.6.7,

$$\begin{array}{cr}
\exists h \in \mathbb{Q}, 0 < h < \min\left\{\alpha, \dfrac{\alpha^n - x}{n\alpha^{n-1}}\right\} & \\
\Downarrow & \text{Lemma 1.6.8} \\
\alpha^n - (\alpha - h)^n \leq hn\alpha^{n-1} & \\
\Downarrow & \\
(\alpha - h)^n \geq \alpha^n - hn\alpha^{n-1} > \alpha^n - (\alpha^n - x) = x & \\
\Downarrow & \\
\forall t \in E, t^n < x < (\alpha - h)^n & \\
\Downarrow & \text{Lemma 1.6.9} \\
\forall t \in E, t < \alpha - h & \\
\Downarrow & \\
\alpha - h \in E^u & \\
\Downarrow & \alpha = \sup E \\
\perp &
\end{array}$$

This implies

$$\exists y \in \mathbb{R}^+, y^n = x$$

Suppose

- $y_1, y_2 \in \mathbb{R}^+$
- $(y_1)^n = (y_2)^n = x$

Assume $y_1 \neq y_2$ (*WLOG* $y_1 < y_2$). Then

$$\begin{array}{cr}
y_1 < y_2 & \\
\Downarrow & \text{Lemma 1.6.9} \\
(y_1)^n < (y_2)^n & \\
\Downarrow & \\
\perp &
\end{array}$$

Therefore,

$$y_1 = y_2$$

This implies

$$\exists! y \in \mathbb{R}^+, y^n = x$$

□

Corollary 1.6.11.

$$\forall x \in [0, \infty), \forall n \in \mathbb{N}, \exists! y \in [0, \infty), y^n = x$$

Such an y is denoted by

$$y = \sqrt[n]{x} = x^{\frac{1}{n}}$$

Proof. Suppose

- $x \in [0, \infty)$
- $n \in \mathbb{N}$

Then

- If $x = 0$, then
  - $0^n = 0$
  - $\forall y \in [0, \infty), y \neq 0 \implies y^n \neq 0$
- Otherwise,

$$\begin{gathered} x \in \mathbb{R}^+ \\ \Downarrow \\ \exists ! y \in \mathbb{R}^+, y^n = x \end{gathered} \qquad \text{Theorem 1.6.10}$$

□

Definition 1.6.12. Suppose

- $y \in [0, 1)$
- $b \in [2, \infty)_{\mathbb{N}}$

A sequence $(a_n)_{n=1}^{\infty}$ in $[0, b-1]_{\mathbb{N}}$ is called a **greedy base** b **expansion** of y if

$$\forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n}$$

Such an $(a_n)_{n=1}^{\infty}$ is denoted by

$$(a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

Theorem 1.6.13.

$$\forall b \in [2, \infty)_{\mathbb{N}}, \forall y \in [0, 1), \exists ! (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}, (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

Proof. Suppose

- $b \in [2, \infty)_{\mathbb{N}}$

- $y \in [0, 1)$

Define

$$\begin{array}{ccc} t_1 := by & T_1 := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_1\} & a_1 := \max T_1 \\ t_2 := b^2 y - ba_1 & T_2 := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_2\} & a_2 := \max T_2 \\ \vdots & \vdots & \vdots \\ t_n := b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} & T_n := \{t \in [0, b-1]_{\mathbb{N}} \mid t \leq t_n\} & a_n := \max T_n \\ \vdots & \vdots & \vdots \end{array}$$

Define

$$\forall n \in \mathbb{N}, P(n) \iff \forall k \in [1, n]_{\mathbb{N}}, 0 \leq t_k < b$$

Then

- **Base case:**

$$\begin{gathered} 0 \leq y < 1 \\ \Downarrow \\ 0 \leq t_1 = by < b \\ \Downarrow \\ P(1) \end{gathered}$$

- **Inductive step:** Suppose $P(n)$. Then

$$\begin{gathered} \forall k \in [1, n]_{\mathbb{N}}, 0 \in T_k \wedge b - 1 \in (T_k)^u \\ \Downarrow \\ \forall k \in [1, n]_{\mathbb{N}}, \exists M \in [0, b-1]_{\mathbb{N}}, M = \max T_k \end{gathered} \qquad \text{Theorem 1.4.5}$$

Therefore, $(a_k)_{k=1}^n$ is well-defined. Then

$$\begin{aligned} t_{n+1} &= b^{n+1} y - \sum_{k=1}^{n} a_k b^{n+1-k} \\ &= b\left(b^n y - \sum_{k=1}^{n} a_k b^{n-k}\right) \\ &= b\left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n\right) \\ &= b(t_n - a_n) \end{aligned}$$

Since $P(n)$,

$$\begin{gathered} a_n \leq t_n < a_n + 1 \\ \Downarrow \\ 0 \leq t_n - a_n < 1 \\ \Downarrow \\ 0 \leq t_{n+1} = b(t_n - a_n) < b \\ \Downarrow \\ P(n+1) \end{gathered}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n)$$

Therefore, $(t_n)_{n=1}^{\infty}$ is well-defined. Then

$$\begin{gathered} \forall n \in \mathbb{N}, 0 \in T_n \land b - 1 \in (T_n)^u \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists M \in [0, b-1]_{\mathbb{N}}, M = \max T_n \end{gathered} \qquad \text{Theorem 1.4.5}$$

Therefore, $(a_n)_{n=1}^{\infty}$ is well-defined. Then

$$\begin{gathered} \forall n \in \mathbb{N}, a_n \leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \Downarrow \\ \forall n \in \mathbb{N}, 0 \leq b^n y - \sum_{k=1}^{n} a_k b^{n-k} < 1 \\ \Downarrow \\ \forall n \in \mathbb{N}, 0 \leq y - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \end{gathered}$$

This implies

$$\exists (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}, (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

Suppose $(a_n)_{n=1}^{\infty} = (c_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$. Define

$$\forall n \in \mathbb{N}, Q(n) \iff \forall k \in [1, n]_{\mathbb{N}}, a_k = c_k$$

Then

- **Base case:** By definition 1.6.12,

$$\begin{gathered} 0 \leq y - \frac{a_1}{b} < \frac{1}{b} \\ \Downarrow \\ \frac{a_1}{b} \leq y < \frac{a_1 + 1}{b} \end{gathered} \qquad \begin{gathered} 0 \leq y - \frac{c_1}{b} < \frac{1}{b} \\ \Downarrow \\ \frac{c_1}{b} \leq y < \frac{c_1 + 1}{b} \end{gathered}$$

Then

$$\begin{array}{c} \frac{a_1}{b} \leq y < \frac{c_1+1}{b} \\ \Downarrow \\ a_1 < c_1 + 1 \\ \Downarrow \\ a_1 \leq c_1 \end{array} \qquad \begin{array}{c} \frac{c_1}{b} \leq y < \frac{a_1+1}{b} \\ \Downarrow \\ c_1 < a_1 + 1 \\ \Downarrow \\ c_1 \leq a_1 \end{array}$$

Therefore,

$$\begin{array}{c} a_1 = c_1 \\ \Downarrow \\ Q(1) \end{array}$$

- **Inductive step:** Suppose $Q(n)$. By definition 1.6.12,

$$\begin{array}{c} 0 \leq y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \end{array} \qquad \begin{array}{c} 0 \leq y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \end{array}$$

Then

$$\begin{array}{c} \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}} \end{array} \qquad \begin{array}{c} \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} \\ \Downarrow \\ \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}} \end{array}$$

Since $Q(n)$,

$$\begin{array}{c} \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} \\ \Downarrow \\ a_{n+1} - c_{n+1} < 1 \\ \Downarrow \\ a_{n+1} \leq c_{n+1} \end{array} \qquad \begin{array}{c} \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} \\ \Downarrow \\ c_{n+1} - a_{n+1} < 1 \\ \Downarrow \\ c_{n+1} \leq a_{n+1} \end{array}$$

Therefore,

$$\begin{array}{c} a_{n+1} = c_{n+1} \\ \Downarrow \\ Q(n+1) \end{array}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, Q(n)$$

This implies

$$\exists!(a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}, (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

□

Theorem 1.6.14.

$$\forall b \in [2, \infty)_{\mathbb{N}}, \forall a, c \in [0, 1), \mathcal{G}_b(a) = \mathcal{G}_b(c) \implies a = c$$

Proof. Suppose

- $b \in [2, \infty)_{\mathbb{N}}$
- $a, c \in [0, 1)$
- $a \neq c$ (*WLOG* $a < c$)

By theorem 1.6.4,

$$\exists n \in \mathbb{N}, \frac{1}{b^n} < c - a$$

By theorem 1.6.13, let

- $\{a_n\}_{n=1}^{\infty} = \mathcal{G}_b(a)$
- $\{c_n\}_{n=1}^{\infty} = \mathcal{G}_b(c)$

By definition 1.6.12,

$$0 \leq a - \sum_{k=1}^{n} \frac{a_k}{b^k} < \frac{1}{b^n} \qquad\qquad 0 \leq c - \sum_{k=1}^{n} \frac{c_k}{b^k} < \frac{1}{b^n}$$

$$\Downarrow \qquad\qquad\qquad\qquad \Downarrow$$

$$\sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} \qquad\qquad c < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n}$$

Therefore,

$$\sum_{k=1}^{n} \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^{n} \frac{c_k}{b^k} + \frac{1}{b^n}$$

$$\Downarrow$$

$$\sum_{k=1}^{n} \frac{a_k}{b^k} < \sum_{k=1}^{n} \frac{c_k}{b^k}$$

$$\Downarrow$$

$$\mathcal{G}_b(a) \neq \mathcal{G}_b(c)$$

□

Theorem 1.6.15.

$$\forall b \in [2, \infty)_{\mathbb{N}}, \forall (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}},$$
$$(\nexists y \in [0,1), (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)) \implies \exists N \in \mathbb{N}, \forall n \in [N, \infty)_{\mathbb{N}}, a_n = b-1$$

Proof. Suppose

- $b \in [2, \infty)_{\mathbb{N}}$
- $(a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}$
- $\forall N \in \mathbb{N}, \exists n \in [N, \infty)_{\mathbb{N}}, a_n \neq b-1$

Define

$$\forall n \in \mathbb{N}, x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

Then

- Suppose
  - $n, m \in \mathbb{N}$
  - $n \leq m$

  Then

  $$\begin{gathered} \forall k \in \mathbb{N}, a_k \geq 0 \\ \Downarrow \\ x_n \leq x_m \end{gathered}$$

- Suppose
  - $n, m \in \mathbb{N}$
  - $n > m$

  Then

  $$\begin{gathered} x_n = x_m + \sum_{k=m+1}^{n} \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^{n} \frac{b-1}{b^k} \\ \Downarrow \\ x_n \leq x_m + \frac{1}{b^m} - \frac{1}{b^n} \end{gathered}$$

Let $N = 1$. Then

$$\begin{gathered}\exists m \in \mathbb{N}, a_m \neq b-1 \\ \Downarrow \\ a_m + 1 \leq b-1\end{gathered}$$

This implies

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m+1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^{m} \frac{b-1}{b^k} = 1 - \frac{1}{b^m} \end{aligned}$$

Then

- $\forall n \in [1, m]_{\mathbb{N}}, x_n \leq x_m$
- $\forall n \in [m+1, \infty)_{\mathbb{N}}, x_n < x_m + \dfrac{1}{b^m} \leq 1 - \dfrac{1}{b^m}$

Therefore,

$$\begin{gathered} 1 - \frac{1}{b^m} \in (\{x_n\}_{n=1}^{\infty})^u \\ \Downarrow \qquad \text{Axiom 1.6.1} \\ \exists y \in \mathbb{R}, y = \sup\{x_n\}_{n=1}^{\infty} \\ \Downarrow \\ y \leq 1 - \frac{1}{b^m} < 1 \\ \Downarrow \qquad 0 \leq x_1 \leq y \\ y \in [0, 1) \end{gathered}$$

Suppose $k \in \mathbb{N}$. Then

$$\begin{gathered} \exists m \in [k+1, \infty)_{\mathbb{N}}, a_m \neq b-1 \\ \Downarrow \\ x_m = x_k + \sum_{n=k+1}^{m} \frac{a_n}{b^n} < x_k + \sum_{n=k+1}^{m} \frac{b-1}{b^n} \\ \Downarrow \\ x_m < x_k + \frac{1}{b^k} - \frac{1}{b^m} \\ \Downarrow \\ x_m + \frac{1}{b^m} < x_k + \frac{1}{b^k} \end{gathered}$$

Therefore,

$$x_m + \frac{1}{b^m} \in (\{x_n\}_{n=1}^{\infty})^u$$
$$\Downarrow \qquad y = \sup\{x_n\}_{n=1}^{\infty}$$
$$x_k \leq y \leq x_m + \frac{1}{b^m}$$
$$\Downarrow$$
$$0 \leq y - x_k < \frac{1}{b^k}$$

Since $k \in \mathbb{N}$ was arbitrary,

$$(a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

□

Theorem 1.6.16.

$$\forall b \in [2,\infty)_{\mathbb{N}}, \forall (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}},$$
$$(\exists N \in \mathbb{N}, \forall n \in [N,\infty)_{\mathbb{N}}, a_n = b-1) \implies \nexists y \in [0,1), (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

Proof. Suppose

- $b \in [2,\infty)_{\mathbb{N}}$
- $(a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} [0, b-1]_{\mathbb{N}}$
- $\exists N \in \mathbb{N}, \forall n \in [N,\infty)_{\mathbb{N}}, a_n = b-1$

Assume

$$\exists y \in [0,1), (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

Define

$$\forall n \in \mathbb{N}, x_n := \sum_{k=1}^{n} \frac{a_k}{b^k}$$

Then

$$\forall n \in [N+1,\infty)_{\mathbb{N}}, x_n = x_N + \sum_{k=N+1}^{n} \frac{b-1}{b^k} = x_N + \frac{1}{b^N} - \frac{1}{b^n}$$
$$\Downarrow$$
$$\forall n \in [N+1,\infty)_{\mathbb{N}}, y - x_n = y - x_N - \frac{1}{b^N} + \frac{1}{b^n} \geq 0$$
$$\Downarrow$$
$$\forall n \in [N+1,\infty)_{\mathbb{N}}, \frac{1}{b^n} \geq x_N + \frac{1}{b^N} - y$$

By definition 1.6.12,

$$
\begin{gathered}
0 \leq y - x_N < \frac{1}{b^N} \\
\Downarrow \\
x_N + \frac{1}{b^N} - y > 0 \\
\Downarrow \qquad \text{Theorem 1.6.4} \\
\exists m \in \mathbb{N}, \frac{1}{b^m} < x_N + \frac{1}{b^N} - y \\
\Downarrow \\
\forall n \in [\max\{m, N\} + 1, \infty)_{\mathbb{N}}, \frac{1}{b^n} < \frac{1}{b^m} < x_N + \frac{1}{b^N} - y \\
\Downarrow \\
\perp
\end{gathered}
$$

Therefore,

$$\nexists y \in [0,1), (a_n)_{n=1}^{\infty} = \mathcal{G}_b(y)$$

□

1.7 The Extended Real Number System

Definition 1.7.1. The extended real number system $\overline{\mathbb{R}}$ is defined by

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}$$

$(\overline{\mathbb{R}}, <)$ is an ordered set with

$$\forall x \in \mathbb{R}, -\infty < x < +\infty$$

Definition 1.7.2. Operations on $\overline{\mathbb{R}}$ are defined by

- $\forall x \in (-\infty, +\infty], x + (+\infty) = (+\infty) + x := +\infty$
- $\forall x \in [-\infty, +\infty), x + (-\infty) = (-\infty) + x := -\infty$
- $\forall x \in (0, +\infty], x \cdot (\pm\infty) = (\pm\infty) \cdot x := \pm\infty$
- $\forall x \in [-\infty, 0), x \cdot (\pm\infty) = (\pm\infty) \cdot x := \mp\infty$
- $\forall x \in \mathbb{R}, \dfrac{x}{\pm\infty} := 0$

Undefined operations in $\overline{\mathbb{R}}$ are

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\dfrac{(\pm\infty)}{(\pm\infty)}, \dfrac{(\pm\infty)}{(\mp\infty)}$
- $\forall x \in \overline{\mathbb{R}}, \dfrac{x}{0}$

Theorem 1.7.3.

(A) $\forall E \subseteq \overline{\mathbb{R}}, \exists x \in \overline{\mathbb{R}}, x = \sup E$

(B) $\forall E \subseteq \overline{\mathbb{R}}, \exists y \in \overline{\mathbb{R}}, y = \inf E$

Proof. Suppose $E \subseteq \overline{\mathbb{R}}$. Then

- If $E = \varnothing$, then

  (A) $\sup E = -\infty$

  (B) $\inf E = +\infty$

- If $E = \{-\infty\}$, then

  (A) $\sup E = -\infty$

  (B) $\inf E = -\infty$

- If $E = \{+\infty\}$, then

  (A) $\sup E = +\infty$

  (B) $\inf E = +\infty$

- Otherwise,

  (A) – If

  $$\exists \beta \in \mathbb{R}, \forall x \in E, x \leq \beta$$

  then

  $$\begin{gathered} E \setminus \{-\infty\} \subseteq \mathbb{R} \\ \Downarrow \qquad \text{Axiom 1.6.1} \\ \exists x \in \mathbb{R}, x = \sup E \setminus \{-\infty\} \\ \Downarrow \\ x = \sup E \end{gathered}$$

- Otherwise,
$$\sup E = +\infty$$

(B)
- If
$$\exists \alpha \in \mathbb{R}, \forall x \in E, \alpha \leq x$$
then
$$\begin{gathered} E \setminus \{+\infty\} \subseteq \mathbb{R} \\ \Downarrow \\ \exists x \in \mathbb{R}, x = \inf E \setminus \{+\infty\} \\ \Downarrow \\ x = \inf E \end{gathered}$$
Theorem 1.4.9
- Otherwise,
$$\inf E = -\infty$$

□

1.8 The Complex Number System

Definition 1.8.1. The complex number system $\mathbb{C}$ is defined by
$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}$$

Definition 1.8.2. The operations on $\mathbb{C}$ are defined by

- $\forall (x, y), (u, v) \in \mathbb{C}, (x, y) + (u, v) := (x + u, y + v)$
- $\forall (x, y), (u, v) \in \mathbb{C}, (x, y) \cdot (u, v) := (xu - yv, xv + yu)$

Remark. One can verify that $(\mathbb{C}, +, \cdot)$ is a field with

- The **additive identity** is $(0, 0)$
- The **multiplicative identity** is $(1, 0)$
- The **additive inverse** of (x, y) is $(-x, -y)$
- The **multiplicative inverse** of $(x, y) \neq (0, 0)$ is
$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2}\right)$$

Definition 1.8.3. The **imaginary unit** i is defined by

$$i := (0, 1) \in \mathbb{C}$$

Remark. There is a **canonical embedding**

$$\mathbb{R} \to \mathbb{C} \text{ such that } x \mapsto (x, 0)$$

Therefore, we identify $\mathbb{R}$ with its image in $\mathbb{C}$ and write

$$\mathbb{R} \subseteq \mathbb{C}$$

Then

$$\begin{aligned} \forall (x, y) \in \mathbb{C}, (x, y) &= (x, 0) + (0, y) \\ &= (x, 0) + (y, 0) \cdot (0, 1) \\ &= x + yi \end{aligned}$$

Definition 1.8.4. Suppose $z = (x, y) \in \mathbb{C}$.

- The **real part** of z is defined by
  - $\Re(z) := x$
  - $\mathrm{Re}(z) := x$
- The **imaginary part** of z is defined by
  - $\Im(z) := y$
  - $\mathrm{Im}(z) := y$
- The **complex conjugate** of z is defined by

$$\overline{z} := (x, -y) = x - yi$$

Definition 1.8.5. Suppose $z = (x, y) \in \mathbb{C}$. The **modulus** of z is defined by

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in [0, \infty)$$

By corollary 1.6.11,

$$\forall z \in \mathbb{C}, \exists! r \in [0, \infty), r = |z|$$

Theorem 1.8.6.

$$\forall n \in \mathbb{N}, \forall \{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subseteq \mathcal{P}(\mathbb{C}), \left|\sum_{k=1}^n a_k\overline{b_k}\right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2$$

This theorem is called the ***Cauchy-Schwarz inequality****.*

Proof. Suppose

- $n \in \mathbb{N}$
- $\{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subseteq \mathcal{P}(\mathbb{C})$

Then

$$\begin{aligned}\forall i, j \in [1, n]_{\mathbb{N}}, |a_ib_j - a_jb_i|^2 &= (a_ib_j - a_jb_i)\overline{(a_ib_j - a_jb_i)} \\ &= |a_i|^2|b_j|^2 + |a_j|^2|b_i|^2 - (a_i\overline{b_i} \cdot \overline{a_j}b_j + \overline{a_i}b_i \cdot a_j\overline{b_j})\end{aligned}$$

Consider

$$\begin{aligned}\sum_{i<j}(|a_i|^2|b_j|^2 + |a_j|^2|b_i|^2) &= \sum_{i<j}(|a_i|^2|b_j|^2) + \sum_{j<i}(|a_i|^2|b_j|^2) \\ &= \sum_{i\neq j}(|a_i|^2|b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2|b_k|^2\end{aligned}$$

Also,

$$\begin{aligned}\sum_{i<j}(a_i\overline{b_i} \cdot \overline{a_j}b_j + \overline{a_i}b_i \cdot a_j\overline{b_j}) &= \sum_{i\neq j}(a_i\overline{b_i} \cdot \overline{a_j}b_j) \\ &= \sum_{i=1}^n a_i\overline{b_i} \cdot \sum_{j=1}^n \overline{a_j}b_j - \sum_{k=1}^n a_k\overline{b_k} \cdot \overline{a_k}b_k \\ &= \left|\sum_{k=1}^n a_k\overline{b_k}\right|^2 - \sum_{k=1}^n |a_k|^2|b_k|^2\end{aligned}$$

Therefore,

$$\sum_{i<j}|a_ib_j - a_jb_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left|\sum_{k=1}^n a_k\overline{b_k}\right|^2 \geq 0$$

$\square$

Corollary 1.8.7.

$$\forall n \in \mathbb{N}, \forall \{a_k\}_{k=1}^n, \{b_k\}_{k=1}^n \subseteq \mathcal{P}(\mathbb{R}), \left(\sum_{k=1}^n a_k b_k\right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2$$

Theorem 1.8.8.

(A) $\forall z, w \in \mathbb{C}, |zw| = |z| \cdot |w|$

(B) $\forall z, w \in \mathbb{C}, |z+w| \leq |z| + |w|$

(C) $\forall z, w \in \mathbb{C}, ||z| - |w|| \leq |z - w|$

Proof. Suppose

- $z = (x, y) \in \mathbb{C}$
- $w = (u, v) \in \mathbb{C}$

Then

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$

(B) By corollary 1.8.7,

$$\begin{aligned} |z+w|^2 &= (x+u)^2 + (y+v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2+y^2} \cdot \sqrt{u^2+v^2} \\ &= (|z| + |w|)^2 \end{aligned}$$

(C) By corollary 1.8.7,

$$\begin{aligned} |z-w|^2 &= (x-u)^2 + (y-v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2+y^2} \cdot \sqrt{u^2+v^2} \\ &= (|z| - |w|)^2 \end{aligned}$$

$\square$

1.9 Euclidean Spaces

Definition 1.9.1. Suppose $n \in \mathbb{N}$. The **Euclidean** n**-space** $\mathbb{R}^n$ is defined by

$$\mathbb{R}^n \coloneqq \{\boldsymbol{x} = (x_1, x_2, \dots, x_n) \mid \forall k \in [1, n]_{\mathbb{N}}, x_k \in \mathbb{R}\}$$

$x_1, x_2, \dots, x_n$ are called the **coordinates** of $\boldsymbol{x}$. The elements of $\mathbb{R}^n$ for $n > 1$ are called **points** and are represented by bold symbols.

Definition 1.9.2. The operations on $\mathbb{R}^n$ are defined by

- $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \boldsymbol{x} + \boldsymbol{y} \coloneqq (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$
- $\forall \alpha \in \mathbb{R}, \forall \boldsymbol{x} \in \mathbb{R}^n, \alpha\boldsymbol{x} \coloneqq (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$

Definition 1.9.3. Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$.

- The **inner product** of $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined by

$$\boldsymbol{x} \cdot \boldsymbol{y} \coloneqq \sum_{k=1}^{n} x_k y_k$$

- The **norm** of $\boldsymbol{x}$ is defined by

$$\|\boldsymbol{x}\| \coloneqq \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{\sum_{k=1}^{n} x_k^2}$$

Theorem 1.9.4.

(A) $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, |\boldsymbol{x} \cdot \boldsymbol{y}| \leq \|\boldsymbol{x}\| \cdot \|\boldsymbol{y}\|$

(B) $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$

(C) $\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, \|\boldsymbol{x} - \boldsymbol{y}\| \geq |\|\boldsymbol{x}\| - \|\boldsymbol{y}\||$

Proof. Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Then

(A) By corollary 1.8.7,

$$|\boldsymbol{x} \cdot \boldsymbol{y}|^2 = \left|\sum_{k=1}^{n} x_k y_k\right|^2 \leq \sum_{k=1}^{n} x_k^2 \cdot \sum_{k=1}^{n} y_k^2 = \|\boldsymbol{x}\|^2 \cdot \|\boldsymbol{y}\|^2$$

(B) By part **(A)**,

$$\begin{aligned}\|\boldsymbol{x}+\boldsymbol{y}\|^2 &= (\boldsymbol{x}+\boldsymbol{y})\cdot(\boldsymbol{x}+\boldsymbol{y})\\ &= \|\boldsymbol{x}\|^2 + 2\boldsymbol{x}\cdot\boldsymbol{y} + \|\boldsymbol{y}\|^2\\ &\le \|\boldsymbol{x}\|^2 + 2|\boldsymbol{x}\cdot\boldsymbol{y}| + \|\boldsymbol{y}\|^2\\ &\le \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\cdot\|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2\\ &= (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2\end{aligned}$$

(C) By part **(B)**,

- If $\|\boldsymbol{x}\| \ge \|\boldsymbol{y}\|$, then

$$\begin{gathered}\|\boldsymbol{x}\| = \|(\boldsymbol{x}-\boldsymbol{y})+\boldsymbol{y}\| \le \|\boldsymbol{x}-\boldsymbol{y}\| + \|\boldsymbol{y}\|\\ \Downarrow\\ \|\boldsymbol{x}\| - \|\boldsymbol{y}\| \le \|\boldsymbol{x}-\boldsymbol{y}\|\end{gathered}$$

- Otherwise,

$$\begin{gathered}\|\boldsymbol{y}\| = \|(\boldsymbol{y}-\boldsymbol{x})+\boldsymbol{x}\| \le \|\boldsymbol{y}-\boldsymbol{x}\| + \|\boldsymbol{x}\|\\ \Downarrow\\ \|\boldsymbol{y}\| - \|\boldsymbol{x}\| \le \|\boldsymbol{x}-\boldsymbol{y}\|\end{gathered}$$

$\square$

Chapter 2

Basic Topology

2.1 Functions

Definition 2.1.1. Suppose a relation G from X to Y satisfies

$$\forall x \in X, \exists! y \in Y, (x, y) \in G$$

Then the ordered triple

$$f \coloneqq (X, Y, G)$$

is called a **function**, where

- X is called the **domain** of f
- Y is called the **codomain** of f
- G is called the **graph** of f

The notation

- $f : X \to Y$ denotes that f is a function from X to Y
- $f(x) = y$ denotes $(x, y) \in G$

Definition 2.1.2. Suppose $f : X \to Y$.

- The **image** of $E \subseteq X$ under f is defined by

$$f(E) \coloneqq \{f(x) \mid x \in E\} \subseteq Y$$

- The **inverse image** of $E \subseteq Y$ under f is defined by

$$f^{-1}(E) \coloneqq \{x \in X \mid f(x) \in E\} \subseteq X$$

Definition 2.1.3. Suppose $f : X \to Y$.

- f is said to be **injective** if

$$\forall x_1, x_2 \in X, f(x_1) = f(x_2) \implies x_1 = x_2$$

  The notation $f : X \hookrightarrow Y$ denotes that f is injective.

- f is said to be **surjective** if

$$\forall y \in Y, \exists x \in X, f(x) = y$$

  or equivalently,

$$f(X) = Y$$

  The notation $f : X \twoheadrightarrow Y$ denotes that f is surjective.

- f is said to be **bijective** if
  - f is injective
  - f is surjective

  or equivalently,

$$\forall y \in Y, \exists! x \in X, f(x) = y$$

  The notation $f : X \xrightarrow{\sim} Y$ denotes that f is bijective.

Definition 2.1.4. Suppose $f : X \xrightarrow{\sim} Y$. The **inverse function** $f^{-1} : Y \to X$ of f is defined by

$$\forall (y, x) \in Y \times X, f^{-1}(y) = x \iff f(x) = y$$

By definition 2.1.3,

$$\begin{gathered} \forall y \in Y, \exists! x \in X, f(x) = y \\ \Downarrow \\ \forall y \in Y, \exists! x \in X, f^{-1}(y) = x \end{gathered}$$

Therefore, f^{-1} is well-defined.

Theorem 2.1.5.

$$\forall f : X \xrightarrow{\sim} Y, f^{-1} : Y \xrightarrow{\sim} X$$

Proof. By definition 2.1.1,

$$\begin{gathered}\forall x \in X, \exists! y \in Y, f(x) = y \\ \Downarrow \\ \forall x \in X, \exists! y \in Y, f^{-1}(y) = x\end{gathered}$$

□

Definition 2.1.6. Suppose $f : X \to Y$. Then

- Suppose $E \subseteq X$. The **restriction** $f|_E : E \to Y$ of f to E is defined by

$$\forall x \in E, f|_E(x) = f(x)$$

- Suppose $f(X) \subseteq F \subseteq Y$. The **corestriction** $f|^F : X \to F$ of f to F is defined by

$$\forall x \in X, f|^F(x) = f(x)$$

- Suppose
  - $E \subseteq X$
  - $f(E) \subseteq F \subseteq Y$

  The **bi-restriction** $f|_E^F : E \to F$ of f to E and F is defined by

$$\forall x \in E, f|_E^F(x) = f(x)$$

Theorem 2.1.7. *Suppose*

- $f : X \to Y$
- $E \subseteq X$
- $f(E) \subseteq F \subseteq Y$

Then

(A) $f : X \hookrightarrow Y \implies f|_E^F : E \hookrightarrow F$

(B) $f(E) = F \implies f|_E^F : E \twoheadrightarrow F$

(C) $f : X \hookrightarrow Y \land f(E) = F \implies f|_E^F : E \xrightarrow{\sim} F$

Proof.

(A) Suppose

- $f : X \hookrightarrow Y$
- $x_1, x_2 \in E$
- $f|_E^F(x_1) = f|_E^F(x_2)$

Then

$$\begin{gathered} f(x_1) = f(x_2) \\ \Downarrow \\ x_1 = x_2 \end{gathered}$$

(B) Suppose $f(E) = F$. Then

$$\begin{gathered} \forall y \in F, \exists x \in E, f(x) = y \\ \Downarrow \\ \forall y \in F, \exists x \in E, f|_E^F(x) = y \end{gathered}$$

(C) follows from **(A)** and **(B)**.

□

Definition 2.1.8. Suppose

- $f : X \to Y$
- $g : Y \to Z$

The **composition** $g \circ f : X \to Z$ of g with f is defined by

$$\forall x \in X, (g \circ f)(x) = g(f(x))$$

Theorem 2.1.9. *Suppose*

- $f : X \to Y$
- $g : Y \to Z$

Then

(A) $f : X \hookrightarrow Y \wedge g : Y \hookrightarrow Z \implies g \circ f : X \hookrightarrow Z$

(B) $f : X \twoheadrightarrow Y \wedge g : Y \twoheadrightarrow Z \implies g \circ f : X \twoheadrightarrow Z$

(C) $f : X \xrightarrow{\sim} Y \wedge g : Y \xrightarrow{\sim} Z \implies g \circ f : X \xrightarrow{\sim} Z$

Proof.

(A) Suppose

- $f : X \hookrightarrow Y$
- $g : Y \hookrightarrow Z$
- $x_1, x_2 \in X$
- $(g \circ f)(x_1) = (g \circ f)(x_2)$

Then

$$f(x_1) = f(x_2)$$
$$\Downarrow$$
$$x_1 = x_2$$

(B) Suppose

- $f : X \twoheadrightarrow Y$
- $g : Y \twoheadrightarrow Z$

Then

- $\forall z \in Z, \exists y \in Y, g(y) = z$
- $\forall y \in Y, \exists x \in X, f(x) = y$

Therefore,

$$\forall z \in Z, \exists x \in X, (g \circ f)(x) = z$$

(C) follows from **(A)** and **(B)**.

$\square$

2.2 Finite Sets

Theorem 2.2.1. *Define* $\sim$ *by*

$$\forall X \forall Y, X \sim Y \iff \exists f : X \xrightarrow{\sim} Y$$

Then $\sim$ *is an equivalence relation.*

Proof. By definition 1.2.2,

- **Reflexivity**: Define $f : X \to X$ by
$$\forall x \in X, f(x) := x$$
Then
  - **Injectivity**: Suppose
    * $x_1, x_2 \in X$
    * $f(x_1) = f(x_2)$

    Then
$$x_1 = x_2$$
  - **Surjectivity**: Suppose $y \in X$. Then
$$f(y) = y$$

  Therefore,
$$\begin{gathered} f : X \xrightarrow{\sim} X \\ \Downarrow \\ X \sim X \end{gathered}$$
- **Symmetry**: Suppose $X \sim Y$. Then
$$\begin{gathered} \exists f : X \xrightarrow{\sim} Y \\ \Downarrow \\ f^{-1} : Y \xrightarrow{\sim} X \\ \Downarrow \\ Y \sim X \end{gathered}$$
Theorem 2.1.5

- **Transitivity**: Suppose
  - $X \sim Y$
  - $Y \sim Z$

  Then
  - $\exists f : X \xrightarrow{\sim} Y$
  - $\exists g : Y \xrightarrow{\sim} Z$

  By theorem 2.1.9,
$$\begin{gathered} g \circ f : X \xrightarrow{\sim} Z \\ \Downarrow \\ X \sim Z \end{gathered}$$

□

Remark. Since there is no **set of all sets** in the underlying set theory, the relation defined in theorem 2.2.1 is a **proper-class** relation rather than a set. Its restriction to any set S becomes an equivalence relation on S.

Definition 2.2.2. Recall the relation $\sim$ defined in theorem 2.2.1. A set X is said to be **finite** if

$$\exists n \in [0, \infty)_{\mathbb{N}}, [1, n]_{\mathbb{N}} \sim X$$

Such an n is called a **cardinality** of X, and is denoted by

$$|X| = n$$

The notation $|X| < \aleph_0$ denotes that X is finite.

Definition 2.2.3. A set X is said to be

- **infinite** if $\neg(|X| < \aleph_0)$
- **countable** if $\mathbb{N} \sim X$
- **at most countable** if $|X| < \aleph_0 \vee \mathbb{N} \sim X$
- **uncountable** if $\neg(|X| < \aleph_0 \vee \mathbb{N} \sim X)$

The notation

- $|X| \geq \aleph_0$ denotes that X is infinite
- $|X| = \aleph_0$ denotes that X is countable
- $|X| \leq \aleph_0$ denotes that X is at most countable
- $|X| > \aleph_0$ denotes that X is uncountable

The order symbols above are not **total orders**; they are just convenient notations.

Lemma 2.2.4.

$$\forall n \in [0, \infty)_{\mathbb{N}}, \forall A \subseteq [1, n]_{\mathbb{N}}, |A| < \aleph_0$$

Proof. Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \iff \forall A \subseteq [1, n]_{\mathbb{N}}, |A| < \aleph_0$$

Then

- **Base case:** Define $f : [1,0]_{\mathbb{N}} \to [1,0]_{\mathbb{N}}$ by

$$\forall k \in [1,0]_{\mathbb{N}}, f(k) := k$$

  Then

$$\begin{gathered} [1,0]_{\mathbb{N}} = \varnothing \\ \Downarrow \\ f : [1,0]_{\mathbb{N}} \xrightarrow{\sim} [1,0]_{\mathbb{N}} \\ \Downarrow \\ |[1,0]_{\mathbb{N}}| < \aleph_0 \end{gathered}$$

  Therefore,

$$\begin{gathered} \forall A \subseteq [1,0]_{\mathbb{N}}, A = [1,0]_{\mathbb{N}} \\ \Downarrow \\ \forall A \subseteq [1,0]_{\mathbb{N}}, |A| < \aleph_0 \\ \Downarrow \\ P(0) \end{gathered}$$

- **Inductive step:** Suppose
  - $P(n)$
  - $A \subseteq [1, n+1]_{\mathbb{N}}$

  Then

  - If $n+1 \notin A$, then

$$\begin{gathered} A \subseteq [1,n]_{\mathbb{N}} \\ \Downarrow \\ |A| < \aleph_0 \end{gathered} \qquad P(n)$$

  - Otherwise, define

$$B := A \setminus \{n+1\} \subseteq [1,n]_{\mathbb{N}}$$

    Since $P(n)$,

$$\begin{gathered} |B| < \aleph_0 \\ \Downarrow \\ \exists m \in [0,\infty)_{\mathbb{N}}, \exists f : [1,m]_{\mathbb{N}} \xrightarrow{\sim} B \end{gathered}$$

    Define $g : [1, m+1]_{\mathbb{N}} \to A$ by

$$\forall k \in [1,m+1]_{\mathbb{N}}, g(k) := \begin{cases} n+1, & \text{if } k = m+1 \\ f(k), & \text{otherwise} \end{cases}$$

    Then

    * **Injectivity**: Suppose

· $k_1, k_2 \in [1, m+1]_{\mathbb{N}}$
· $g(k_1) = g(k_2)$

Then

· If $k_1 = m+1$, then

$$\begin{array}{cr} g(k_1) = n+1 & \\ \Downarrow & \\ g(k_2) = n+1 & \\ \Downarrow & n+1 \notin B \\ k_2 = m+1 & \end{array}$$

· Otherwise,

$$\begin{array}{cr} k_1 \in [1, m]_{\mathbb{N}} & \\ \Downarrow & \\ g(k_1) = f(k_1) & \\ \Downarrow & \\ g(k_2) = f(k_1) & \\ \Downarrow & f(k_1) \neq n+1 \\ g(k_2) = f(k_2) = f(k_1) & \\ \Downarrow & f : [1, m]_{\mathbb{N}} \xrightarrow{\sim} B \\ k_1 = k_2 & \end{array}$$

$*$ **Surjectivity**: Suppose $a \in A$.

· If $a = n+1$, then

$$g(m+1) = a$$

· Otherwise,

$$\begin{array}{cr} a \in B & \\ \Downarrow & f : [1, m]_{\mathbb{N}} \xrightarrow{\sim} B \\ \exists k \in [1, m]_{\mathbb{N}}, f(k) = a & \\ \Downarrow & \\ g(k) = f(k) = a & \end{array}$$

Therefore,

$$\begin{array}{c} g : [1, m+1]_{\mathbb{N}} \xrightarrow{\sim} A \\ \Downarrow \\ |A| < \aleph_0 \end{array}$$

Therefore, $P(n+1)$.

By theorem 1.1.2,

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n)$$

□

Theorem 2.2.5.

$$\forall X \forall Y, |X| < \aleph_0 \wedge Y \subseteq X \implies |Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $Y \subseteq X$

Then

$$\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

Define

$$Z := f^{-1}(Y) \subseteq [1, n]_{\mathbb{N}}$$

By lemma 2.2.4,

$$|Z| < \aleph_0$$

By theorem 2.1.7,

$$\begin{gathered} f|_Z^Y : Z \xrightarrow{\sim} Y \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.1}$$

□

Corollary 2.2.6.

$$\forall X \forall Y, |Y| \geq \aleph_0 \wedge Y \subseteq X \implies |X| \geq \aleph_0$$

Proof. Suppose

- $|Y| \geq \aleph_0$
- $Y \subseteq X$

Assume $|X| < \aleph_0$. By theorem 2.2.5,

$$\begin{gathered} |Y| < \aleph_0 \\ \Downarrow \\ \bot \end{gathered}$$

Therefore,

$$|X| \geq \aleph_0$$

□

Lemma 2.2.7.

$$\forall Y, (\exists n \in [0,\infty)_{\mathbb{N}}, \exists f : [1,n]_{\mathbb{N}} \twoheadrightarrow Y) \implies |Y| < \aleph_0$$

Proof. Suppose

$$\exists n \in [0,\infty)_{\mathbb{N}}, \exists f : [1,n]_{\mathbb{N}} \twoheadrightarrow Y$$

Define

$$\forall y \in Y, N_y := f^{-1}(\{y\}) \neq \varnothing$$

By theorem 1.1.4,

$$\forall y \in Y, \exists n_y \in N_y, n_y = \min N_y$$

Define

- $M := \{n_y \mid y \in Y\} \subseteq [1,n]_{\mathbb{N}}$
- $g : M \to Y$ by $g \equiv f|_M^Y$

Then

- **Injectivity**: Suppose
  - $n_{y_1}, n_{y_2} \in M$
  - $g(n_{y_1}) = g(n_{y_2})$

  Then

$$\begin{gathered} f(n_{y_1}) = f(n_{y_2}) \\ \Downarrow \\ y_1 = y_2 \\ \Downarrow \\ N_{y_1} = N_{y_2} \\ \Downarrow \\ n_{y_1} = n_{y_2} \end{gathered}$$

- **Surjectivity**: Suppose $y \in Y$. Then

$$\begin{gathered} \exists n_y \in N_y, n_y = \min N_y \\ \Downarrow \\ g(n_y) = y \end{gathered}$$

Therefore,

$$g : M \xrightarrow{\sim} Y$$

By lemma 2.2.4,

$$\begin{gathered} |M| < \aleph_0 \\ \Downarrow \\ |Y| < \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.1}$$

$\square$

Theorem 2.2.8.

$$\forall X \forall Y, |X| < \aleph_0 \land \exists f : X \twoheadrightarrow Y \implies |Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $\exists f : X \twoheadrightarrow Y$

Then

$$\begin{array}{cr} \exists n \in [0, \infty)_{\mathbb{N}}, \exists g : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X & \\ \Downarrow & \text{Theorem 2.1.9} \\ f \circ g : [1, n]_{\mathbb{N}} \twoheadrightarrow Y & \\ \Downarrow & \text{Lemma 2.2.7} \\ |Y| < \aleph_0 & \end{array}$$

□

Remark. Suppose

- $|X| < \aleph_0$
- $X \neq \varnothing$

Then

$$\exists n \in \mathbb{N}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$$

This implies

$$X = \{f(1), f(2), \ldots, f(n)\}$$

where

$$\forall i, j \in [1, n]_{\mathbb{N}}, i \neq j \implies f(i) \neq f(j)$$

Theorem 2.2.9.

$$\forall (X, <), |X| < \aleph_0 \land X \neq \varnothing \implies \exists m, M \in X, m = \min X \land M = \max X$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) \iff \forall (X, <), |X| = n \implies \exists m, M \in X, m = \min X \land M = \max X$$

Then

- **Base case:** Suppose $X = \{a_1\}$. Then
  - $\forall x \in X, a_1 \leq x$
  - $\forall x \in X, x \leq a_1$

  Therefore, $P(1)$.

- **Inductive step:** Suppose
  - $P(n)$
  - $X = \{a_1, a_2, \ldots, a_{n+1}\}$
  - $Y = \{a_1, a_2, \ldots, a_n\} \subseteq X$

  Since $P(n)$,
  - $\exists k \in [1, n]_{\mathbb{N}}, \forall y \in Y, a_k \leq y$
  - $\exists K \in [1, n]_{\mathbb{N}}, \forall y \in Y, y \leq a_K$

  Then
  - $\forall x \in X, \text{min}\{a_k, a_{n+1}\} \leq x$
  - $\forall x \in X, x \leq \text{max}\{a_K, a_{n+1}\}$

  Therefore, $P(n+1)$.

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n)$$

$\square$

Lemma 2.2.10.

(A) $\forall n, m \in [0, \infty)_{\mathbb{N}}, (\exists f : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}) \implies n \leq m$

(B) $\forall n, m \in [0, \infty)_{\mathbb{N}}, (\exists f : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}) \implies n \geq m \geq 1 \vee n = m = 0$

(C) $\forall n, m \in [0, \infty)_{\mathbb{N}}, \left(\exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}\right) \implies n = m$

Proof. Suppose $n, m \in [0, \infty)_{\mathbb{N}}$. Then

(A) Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \iff \forall m \in [0, \infty)_{\mathbb{N}}, (\exists f_n : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}) \implies n \leq m$$

Then

- **Base case:**
$$\begin{gathered}\forall m \in [0,\infty)_{\mathbb{N}}, 0 \leq m \\ \Downarrow \\ P(0)\end{gathered}$$
- **Inductive step:** Suppose
  - $P(n)$
  - $\exists f_{n+1} : [1, n+1]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}$

  Then $m \geq 1$. Let
$$f_{n+1}(n+1) = x \in [1, m]_{\mathbb{N}}$$
  Define $g_m : [1, m]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by
$$\forall y \in [1, m]_{\mathbb{N}}, g_m(y) := \begin{cases} x, & \text{if } y = m \\ m, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$
  Then
  - **Injectivity**: Suppose
    * $y_1, y_2 \in [1, m]_{\mathbb{N}}$
    * $g_m(y_1) = g_m(y_2)$

    Then
    * If $y_1 = m$, then
$$\begin{gathered}g_m(y_1) = x \\ \Downarrow \\ g_m(y_2) = x \\ \Downarrow \\ y_2 = m\end{gathered}$$
    * If $y_1 = x$, then
$$\begin{gathered}g_m(y_1) = m \\ \Downarrow \\ g_m(y_2) = m \\ \Downarrow \\ y_2 = x\end{gathered}$$
    * Otherwise,
$$\begin{aligned}&g_m(y_1) = y_1 \\ &\Downarrow \\ &g_m(y_2) = y_1 \\ &\Downarrow && y_1 \neq m \wedge y_1 \neq x \\ &g_m(y_2) = y_2 \\ &\Downarrow \\ &y_2 = y_1\end{aligned}$$

- **Surjectivity**: Suppose $k \in [1, m]_{\mathbb{N}}$. Then
  - If $k = m$, then
$$g_m(x) = k$$
  - If $k = x$, then
$$g_m(m) = k$$
  - Otherwise,
$$g_m(k) = k$$

Therefore,
$$g_m : [1, m]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}$$

Also,

- $g_m \circ f_{n+1}(n + 1) = m$
- $(g_m \circ f_{n+1})([1, n]_{\mathbb{N}}) \subseteq [1, m - 1]_{\mathbb{N}}$

Define $f_n : [1, n]_{\mathbb{N}} \to [1, m - 1]_{\mathbb{N}}$ by
$$f_n \equiv (g_m \circ f_{n+1})|_{[1,n]_{\mathbb{N}}}^{[1,m-1]_{\mathbb{N}}}$$

By theorems 2.1.7 and 2.1.9,
$$\begin{gathered} f_n : [1, n]_{\mathbb{N}} \hookrightarrow [1, m - 1]_{\mathbb{N}} \\ \Downarrow \\ n \leq m - 1 \\ \Downarrow \\ n + 1 \leq m \end{gathered} \qquad P(n)$$

Therefore, $P(n + 1)$.

By theorem 1.1.2,
$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n)$$

(B) Define
$$\forall m \in [0, \infty)_{\mathbb{N}}, Q(m) \iff \forall n \in [0, \infty)_{\mathbb{N}},$$
$$(\exists f_m : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}) \implies n \geq m \geq 1 \lor n = m = 0$$

Then

- **Base case:**

- ($m = 0$) Suppose $n \in [0, \infty)_{\mathbb{N}}$.

$$\begin{array}{c} \neg(m \geq 1) \\ \Downarrow \\ n \geq m \geq 1 \vee n = m = 0 \iff n = m = 0 \iff n = 0 \end{array}$$

  Then

$$\begin{array}{c} n \neq 0 \implies \nexists f_m : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ Q(0) \end{array}$$

- ($m = 1$) Suppose $n \in [0, \infty)_{\mathbb{N}}$.

$$\begin{array}{c} \neg(m = 0) \\ \Downarrow \\ n \geq m \geq 1 \vee n = m = 0 \iff n \geq m \geq 1 \iff n \geq 1 \end{array}$$

  Then

$$\begin{array}{c} n < 1 \implies \nexists f_m : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ Q(1) \end{array}$$

- **Inductive step:** Let $m \geq 1$. Suppose
  - $Q(m)$
  - $\exists f_{m+1} : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m + 1]_{\mathbb{N}}$

  Then $n \geq 1$. Let

$$f_{m+1}(n) = x \in [1, m + 1]_{\mathbb{N}}$$

  Define $g_{m+1} : [1, m + 1]_{\mathbb{N}} \to [1, m + 1]_{\mathbb{N}}$ by

$$\forall y \in [1, m + 1]_{\mathbb{N}}, g_{m+1}(y) \coloneqq \begin{cases} x, & \text{if } y = m + 1 \\ m + 1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

  Then
  - **Injectivity**: Suppose
    * $y_1, y_2 \in [1, m + 1]_{\mathbb{N}}$
    * $g_{m+1}(y_1) = g_{m+1}(y_2)$

    Then

* If $y_1 = m + 1$, then

$$\begin{gathered} g_{m+1}(y_1) = x \\ \Downarrow \\ g_{m+1}(y_2) = x \\ \Downarrow \\ y_2 = m + 1 \end{gathered}$$

* If $y_1 = x$, then

$$\begin{gathered} g_{m+1}(y_1) = m + 1 \\ \Downarrow \\ g_{m+1}(y_2) = m + 1 \\ \Downarrow \\ y_2 = x \end{gathered}$$

* Otherwise,

$$\begin{gathered} g_{m+1}(y_1) = y_1 \\ \Downarrow \\ g_{m+1}(y_2) = y_1 \\ \Downarrow \\ g_{m+1}(y_2) = y_2 \\ \Downarrow \\ y_2 = y_1 \end{gathered} \qquad y_1 \neq m + 1 \wedge y_1 \neq x$$

- **Surjectivity**: Suppose $k \in [1, m + 1]_{\mathbb{N}}$. Then
  * If $k = m + 1$, then

$$g_{m+1}(x) = k$$

  * If $k = x$, then

$$g_{m+1}(m + 1) = k$$

  * Otherwise,

$$g_{m+1}(k) = k$$

Therefore,

$$g_{m+1} : [1, m + 1]_{\mathbb{N}} \xrightarrow{\sim} [1, m + 1]_{\mathbb{N}}$$

Also,

$$(g_{m+1} \circ f_{m+1})(n) = m + 1$$

Define

$$K_{m+1} \coloneqq (g_{m+1} \circ f_{m+1})^{-1}(\{m + 1\}) \subseteq [1, n]_{\mathbb{N}}$$

Then

$$(g_{m+1} \circ f_{m+1})([1, n-1]_{\mathbb{N}} \setminus K_{m+1}) = [1, m]_{\mathbb{N}}$$

Define $f_m : [1, n-1]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$\forall y \in [1, n-1]_{\mathbb{N}}, f_m(y) := \begin{cases} m, & \text{if } y \in K_{m+1} \\ (g_{m+1} \circ f_{m+1})(y), & \text{otherwise} \end{cases}$$

By theorems 2.1.7 and 2.1.9,

$$\begin{gathered} (g_{m+1} \circ f_{m+1})|^{[1,m]_{\mathbb{N}}}_{[1,n-1]_{\mathbb{N}} \setminus K_{m+1}} : [1, n-1]_{\mathbb{N}} \setminus K_{m+1} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ f_m : [1, n-1]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n - 1 \geq m \\ \Downarrow \\ n \geq m + 1 \geq 1 \end{gathered} \qquad Q(m)$$

Therefore, $Q(m+1)$.

By theorem 1.1.2,

$$\forall m \in [0, \infty)_{\mathbb{N}}, Q(m)$$

(C) follows from **(A)** and **(B)**.

□

Corollary 2.2.11.

$$\forall X, |X| < \aleph_0 \implies \exists! n \in [0, \infty)_{\mathbb{N}}, |X| = n$$

Proof. Suppose $|X| < \aleph_0$. Then

$$\exists n \in [0, \infty)_{\mathbb{N}}, |X| = n$$

Suppose $|X| = m \in [0, \infty)_{\mathbb{N}}$. Then

$$\begin{gathered} [1, n]_{\mathbb{N}} \sim X \wedge [1, m]_{\mathbb{N}} \sim X \\ \Downarrow \quad \text{Theorem 2.2.1} \\ [1, n]_{\mathbb{N}} \sim [1, m]_{\mathbb{N}} \\ \Downarrow \quad \text{Lemma 2.2.10} \\ n = m \end{gathered}$$

□

Lemma 2.2.12.

(A) $\forall n, m \in [0,\infty)_{\mathbb{N}}, n \leq m \implies \exists f : [1,n]_{\mathbb{N}} \hookrightarrow [1,m]_{\mathbb{N}}$

(B) $\forall n, m \in [0,\infty)_{\mathbb{N}}, n \geq m \geq 1 \vee n = m = 0 \implies \exists f : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}}$

(C) $\forall n, m \in [0,\infty)_{\mathbb{N}}, n = m \implies \exists f : [1,n]_{\mathbb{N}} \xrightarrow{\sim} [1,m]_{\mathbb{N}}$

Proof. Suppose $n, m \in [0,\infty)_{\mathbb{N}}$. Then

(A) Suppose $n \leq m$. Define $f : [1,n]_{\mathbb{N}} \to [1,m]_{\mathbb{N}}$ by

$$\forall i \in [1,n]_{\mathbb{N}}, f(i) := i$$

Suppose

- $i, j \in [1,n]_{\mathbb{N}}$
- $f(i) = f(j)$

Then

$$\begin{gathered} f(i) = f(j) \\ \Downarrow \\ i = j \end{gathered}$$

(B)

- Suppose $n = m = 0$. Define $f : [1,n]_{\mathbb{N}} \to [1,m]_{\mathbb{N}}$ by

$$\forall i \in [1,n]_{\mathbb{N}}, f(i) := i$$

Then

$$\begin{gathered} [1,m]_{\mathbb{N}} = \varnothing \\ \Downarrow \\ f : [1,n]_{\mathbb{N}} \twoheadrightarrow [1,m]_{\mathbb{N}} \end{gathered}$$

- Suppose $n \geq m \geq 1$. Define $f : [1,n]_{\mathbb{N}} \to [1,m]_{\mathbb{N}}$ by

$$\forall i \in [1,n]_{\mathbb{N}}, f(i) := \begin{cases} i, & \text{if } i < m \\ m, & \text{otherwise} \end{cases}$$

Suppose $j \in [1,m]_{\mathbb{N}}$. Then

  - If $j = m$, then

$$f(n) = j$$

  - Otherwise,

$$f(j) = j$$

(C) Suppose $n = m$. Define $f : [1, n]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$\forall i \in [1, n]_{\mathbb{N}}, f(i) := i$$

Then

- **Injectivity**: Suppose
  - $i, j \in [1, m]_{\mathbb{N}}$
  - $f(i) = f(j)$

  Then
$$i = j$$
- **Surjectivity**: Suppose $j \in [1, m]_{\mathbb{N}}$. Then
$$f(j) = j$$

Therefore,
$$f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}}$$

□

Theorem 2.2.13.

(A) $\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies (|X| \leq |Y| \iff \exists f : X \hookrightarrow Y)$

(B) $\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies$
$$(|X| \geq |Y| > 0 \vee |X| = |Y| = 0 \iff \exists f : X \twoheadrightarrow Y)$$

(C) $\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies \Big(|X| = |Y| \iff \exists f : X \xrightarrow{\sim} Y\Big)$

Proof. Suppose

- $|X| = n \in [0, \infty)_{\mathbb{N}}$
- $|Y| = m \in [0, \infty)_{\mathbb{N}}$

Then

- $\exists f_X : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists f_Y : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$

This implies

- ($\Longleftarrow$) Suppose $f : X \to Y$. Define $F : [1, n]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$ by

$$F \equiv (f_Y)^{-1} \circ f \circ f_X$$

By theorems 2.1.5 and 2.1.9,

- $f : X \hookrightarrow Y \implies F : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}}$
- $f : X \twoheadrightarrow Y \implies F : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}}$

By lemma 2.2.10,

(A) Suppose $\exists f : X \hookrightarrow Y$. Then

$$\begin{gathered} \exists F : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n \leq m \end{gathered}$$

(B) Suppose $\exists f : X \twoheadrightarrow Y$. Then

$$\begin{gathered} \exists F : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ n \geq m \geq 1 \vee n = m = 0 \end{gathered}$$

(C) Suppose $\exists f : X \xrightarrow{\sim} Y$. Then

$$\begin{gathered} \exists F : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}} \\ \Downarrow \\ n = m \end{gathered}$$

- ($\Longrightarrow$) Suppose $G : [1, n]_{\mathbb{N}} \to [1, m]_{\mathbb{N}}$. Define $f : X \to Y$ by

$$f \equiv f_Y \circ G \circ (f_X)^{-1}$$

By theorems 2.1.5 and 2.1.9,

- $G : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}} \implies f : X \hookrightarrow Y$
- $G : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \implies f : X \twoheadrightarrow Y$

By lemma 2.2.12,

(A) Suppose $n \leq m$. Then

$$\begin{gathered} \exists G : [1, n]_{\mathbb{N}} \hookrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \hookrightarrow Y \end{gathered}$$

(B) Suppose $n \geq m \geq 1 \vee n = m = 0$. Then

$$\begin{gathered}\exists G : [1, n]_{\mathbb{N}} \twoheadrightarrow [1, m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \twoheadrightarrow Y\end{gathered}$$

(C) Suppose $n = m$. Then

$$\begin{gathered}\exists G : [1, n]_{\mathbb{N}} \xrightarrow{\sim} [1, m]_{\mathbb{N}} \\ \Downarrow \\ \exists f : X \xrightarrow{\sim} Y\end{gathered}$$

$\square$

Lemma 2.2.14.

$$\forall X \forall Y, X \cap Y = \varnothing \wedge |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Suppose

- $X \cap Y = \varnothing$
- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

- $\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists m \in [0, \infty)_{\mathbb{N}}, \exists g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$

Define $h : [1, m + n]_{\mathbb{N}} \to X \cup Y$ by

$$\forall i \in [1, m + n]_{\mathbb{N}}, h(i) := \begin{cases} f(i), & \text{if } i \leq n \\ g(i - n), & \text{otherwise} \end{cases}$$

Then

- **Injectivity**: Suppose
  - $i, j \in [1, m + n]_{\mathbb{N}}$
  - $h(i) = h(j)$

  Then

- If $i \leq n$, then

$$\begin{array}{c} h(i) = f(i) \\ \Downarrow \\ f(i) = h(j) \in X \\ \Downarrow \\ j \leq n \\ \Downarrow \\ h(j) = f(j) \\ \Downarrow \\ f(i) = f(j) \\ \Downarrow \\ i = j \end{array} \qquad \begin{array}{r} \\ \\ \\ X \cap Y = \varnothing \\ \\ \\ \\ \\ \\ f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X \\ \\ \end{array}$$

- Otherwise,

$$\begin{array}{c} h(i) = g(i - n) \\ \Downarrow \\ g(i - n) = h(j) \in Y \\ \Downarrow \\ j > n \\ \Downarrow \\ h(j) = g(j - n) \\ \Downarrow \\ g(i - n) = g(j - n) \\ \Downarrow \\ i - n = j - n \\ \Downarrow \\ i = j \end{array} \qquad \begin{array}{r} \\ \\ \\ X \cap Y = \varnothing \\ \\ \\ \\ \\ \\ g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ \\ \\ \\ \end{array}$$

- **Surjectivity**: Suppose $k \in X \cup Y$. Then
  - If $k \in X$, then

$$\begin{array}{c} k \in X \\ \Downarrow \\ \exists i \in [1, n]_{\mathbb{N}}, f(i) = k \\ \Downarrow \\ h(i) = k \end{array} \qquad \begin{array}{r} \\ f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X \\ \\ \\ \\ \end{array}$$

  - Otherwise,

$$\begin{array}{c} k \in Y \\ \Downarrow \\ \exists j \in [1, m]_{\mathbb{N}}, g(j) = k \\ \Downarrow \\ h(j + n) = k \end{array} \qquad \begin{array}{r} \\ g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ \\ \\ \\ \end{array}$$

Therefore,

$$h : [1, m+n]_{\mathbb{N}} \xrightarrow{\sim} X \cup Y$$
$$\Downarrow$$
$$|X \cup Y| < \aleph_0$$

□

Lemma 2.2.15.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \cup Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Define

- $A := X \setminus Y \subseteq X$
- $B := Y \setminus X \subseteq Y$
- $C := X \cap Y \subseteq X$

By theorem 2.2.5,

- $|A| < \aleph_0$
- $|B| < \aleph_0$
- $|C| < \aleph_0$

Also,

- $A \cap B = B \cap C = C \cap A = \varnothing$
- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$
- $X \cup Y = (X \setminus Y) \cup (Y \setminus X) \cup (X \cap Y) = A \cup B \cup C$

By lemma 2.2.14,

$$|A \cup B| < \aleph_0$$
$$\Downarrow$$
$$|(A \cup B) \cup C| < \aleph_0$$
$$\Downarrow$$
$$|X \cup Y| < \aleph_0$$

□

Theorem 2.2.16.

$$\forall n \in [0,\infty)_{\mathbb{N}}, \forall \{X_k\}_{k=1}^{n}, (\forall k \in [1,n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$$

Proof. Define

$$\forall n \in [0,\infty)_{\mathbb{N}}, P(n) \iff \forall \{X_k\}_{k=1}^{n}, (\forall k \in [1,n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$$

Then

- **Base case:**

$$\begin{gathered}\bigcup_{k=1}^{0} X_k = \varnothing \\ \Downarrow \\ \left|\bigcup_{k=1}^{0} X_k\right| < \aleph_0 \\ \Downarrow \\ P(0)\end{gathered}$$

- **Inductive step:** Suppose
  - $P(n)$
  - $\forall k \in [1,n+1]_{\mathbb{N}}, |X_k| < \aleph_0$

  Since $P(n)$,

$$\left|\bigcup_{k=1}^{n} X_k\right| < \aleph_0$$

  Then

$$\begin{gathered}\bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^{n} X_k\right) \cup X_{n+1} \\ \Downarrow \qquad \text{Lemma 2.2.15} \\ \left|\bigcup_{k=1}^{n+1} X_k\right| < \aleph_0 \\ \Downarrow \\ P(n+1)\end{gathered}$$

By theorem 1.1.2,

$$\forall n \in [0,\infty)_{\mathbb{N}}, P(n)$$

□

Lemma 2.2.17.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| < \aleph_0 \implies |X \times Y| < \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| < \aleph_0$

Then

- $\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X$
- $\exists m \in [0, \infty)_{\mathbb{N}}, \exists g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y$

This implies

- If $n = 0$ or $m = 0$, then
$$\begin{gathered} X \times Y = \varnothing \\ \Downarrow \\ |X \times Y| < \aleph_0 \end{gathered}$$
- Otherwise, define $h : [1, nm]_{\mathbb{N}} \to X \times Y$ by
$$\forall k \in [1, nm]_{\mathbb{N}}, h(k) := \left(f\left(\left\lceil \frac{k}{m} \right\rceil \right), g\left(k - m\left(\left\lceil \frac{k}{m} \right\rceil - 1 \right) \right) \right)$$
  Suppose
  - $k \in [1, nm]_{\mathbb{N}}$
  - $j_k = \left\lceil \dfrac{k}{m} \right\rceil \in \mathbb{Z}$

  Then
  - Since $k \in [1, nm]_{\mathbb{N}}$,
$$\begin{gathered} 1 \leq k \leq nm \\ \Downarrow \\ 1 \leq \left\lceil \frac{k}{m} \right\rceil \leq n \\ \Downarrow \\ j_k \in [1, n]_{\mathbb{N}} \end{gathered}$$

– By definition 1.6.6,

$$\begin{gathered} j_k = \min\left\{ i \in \mathbb{Z} \;\middle|\; i \geq \frac{k}{m} \right\} \\ \Downarrow \\ j_k - 1 < \frac{k}{m} \leq j_k \\ \Downarrow \\ 0 < k - m(j_k - 1) \leq m \\ \Downarrow \\ k - m(j_k - 1) \in [1, m]_{\mathbb{N}} \end{gathered}$$

Therefore, h is well-defined. Then

– **Injectivity**: Suppose
  * $k_1, k_2 \in [1, nm]_{\mathbb{N}}$
  * $h(k_1) = h(k_2)$

  Then

$$\begin{gathered} f(j_{k_1}) = f(j_{k_2}) \\ \Downarrow \qquad\qquad f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X \\ j_{k_1} = j_{k_2} \end{gathered}$$

  Therefore,

$$\begin{gathered} g(k_1 - m(j_{k_1} - 1)) = g(k_2 - m(j_{k_2} - 1)) \\ \Downarrow \qquad\qquad g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ k_1 - m(j_{k_1} - 1) = k_2 - m(j_{k_2} - 1) \\ \Downarrow \qquad\qquad j_{k_1} = j_{k_2} \\ k_1 = k_2 \end{gathered}$$

– **Surjectivity**: Suppose $(x, y) \in X \times Y$. Then

$$\begin{gathered} x \in X \\ \Downarrow \qquad\qquad f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} X \\ \exists j \in [1, n]_{\mathbb{N}}, f(j) = x \end{gathered}$$

  Also,

$$\begin{gathered} y \in Y \\ \Downarrow \qquad\qquad g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} Y \\ \exists i \in [1, m]_{\mathbb{N}}, g(i) = y \end{gathered}$$

  Define

$$k := m(j - 1) + i \in [1, nm]_{\mathbb{N}}$$

  Then

$$h(k) = (f(j), g(i)) = (x, y)$$

Therefore,

$$\begin{gathered} h : [1, nm]_{\mathbb{N}} \xrightarrow{\sim} X \times Y \\ \Downarrow \\ |X \times Y| < \aleph_0 \end{gathered}$$

□

Lemma 2.2.18.

$$\forall n, m \in [0, \infty)_{\mathbb{N}}, \forall \{X_k\}_{k=1}^{n+m}, \prod_{k=1}^{n+m} X_k \sim \left(\prod_{k=1}^{n} X_k \right) \times \left(\prod_{k=n+1}^{n+m} X_k \right)$$

Proof. Suppose $n, m \in [0, \infty)_{\mathbb{N}}$. Define

$$h : \prod_{k=1}^{n+m} X_k \to \left(\prod_{k=1}^{n} X_k \right) \times \left(\prod_{k=n+1}^{n+m} X_k \right)$$

by

$$h(x_1, x_2, \ldots, x_{n+m}) = ((x_1, x_2, \ldots, x_n), (x_{n+1}, x_{n+2}, \ldots, x_{n+m}))$$

Then

- **Injectivity**: Suppose
  - $(x_1, x_2, \ldots, x_{n+m}), (y_1, y_2, \ldots, y_{n+m}) \in \prod_{k=1}^{n+m} X_k$
  - $h(x_1, x_2, \ldots, x_{n+m}) = h(y_1, y_2, \ldots, y_{n+m})$

  Then

  $$\begin{aligned} &((x_1, x_2, \ldots, x_n), (x_{n+1}, x_{n+2}, \ldots, x_{n+m})) \\ &= ((y_1, y_2, \ldots, y_n), (y_{n+1}, y_{n+2}, \ldots, y_{n+m})) \end{aligned}$$

  This implies

  - $(x_1, x_2, \ldots, x_n) = (y_1, y_2, \ldots, y_n)$
  - $(x_{n+1}, x_{n+2}, \ldots, x_{n+m}) = (y_{n+1}, y_{n+2}, \ldots, y_{n+m})$

  Therefore,

  $$(x_1, x_2, \ldots, x_{n+m}) = (y_1, y_2, \ldots, y_{n+m})$$

- **Surjectivity**: Suppose

$$((x_1, x_2, \ldots, x_n), (x_{n+1}, x_{n+2}, \ldots, x_{n+m})) \in \left(\prod_{k=1}^{n} X_k\right) \times \left(\prod_{k=n+1}^{n+m} X_k\right)$$

Then

$$h(x_1, x_2, \ldots, x_{n+m}) = ((x_1, x_2, \ldots, x_n), (x_{n+1}, x_{n+2}, \ldots, x_{n+m}))$$

Therefore,

$$\begin{gathered} h : \prod_{k=1}^{n+m} X_k \xrightarrow{\sim} \left(\prod_{k=1}^{n} X_k\right) \times \left(\prod_{k=n+1}^{n+m} X_k\right) \\ \Downarrow \\ \prod_{k=1}^{n+m} X_k \sim \left(\prod_{k=1}^{n} X_k\right) \times \left(\prod_{k=n+1}^{n+m} X_k\right) \end{gathered}$$

$\square$

Theorem 2.2.19.

$$\forall n \in [0, \infty)_{\mathbb{N}}, \forall \{X_k\}_{k=1}^{n}, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| < \aleph_0$$

Proof. Define

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n) \iff \forall \{X_k\}_{k=1}^{n}, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| < \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| < \aleph_0$$

Then

- **Base case:**

$$\begin{gathered} \prod_{k=1}^{0} X_k = \{()\} \\ \Downarrow \\ \left|\prod_{k=1}^{0} X_k\right| < \aleph_0 \\ \Downarrow \\ P(0) \end{gathered}$$

- **Inductive step:** Suppose

- $P(n)$
- $\forall k \in [1, n+1]_{\mathbb{N}}, |X_k| < \aleph_0$

Since $P(n)$,

$$\left| \prod_{k=1}^{n} X_k \right| < \aleph_0$$

By lemma 2.2.18,

$$\begin{gathered} \prod_{k=1}^{n+1} X_k \sim \left(\prod_{k=1}^{n} X_k \right) \times X_{n+1} \\ \Downarrow \\ \left| \prod_{k=1}^{n+1} X_k \right| < \aleph_0 \\ \Downarrow \\ P(n+1) \end{gathered} \qquad \text{Lemma 2.2.17}$$

By theorem 1.1.2,

$$\forall n \in [0, \infty)_{\mathbb{N}}, P(n)$$

□

Theorem 2.2.20.

$$\forall X, \nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

*This theorem is called **Cantor's theorem**.*

Proof. Assume

$$\exists X, \exists f : X \twoheadrightarrow \mathcal{P}(X)$$

Define

$$S := \{x \in X \mid x \notin f(x)\} \in \mathcal{P}(X)$$

Then

$$\exists x \in X, f(x) = S$$

This implies

- If $x \in S$, then

$$\begin{gathered} x \notin f(x) = S \\ \Downarrow \\ \perp \end{gathered}$$

- Otherwise,

$$\begin{gathered} x \in f(x) = S \\ \Downarrow \\ \bot \end{gathered}$$

Therefore,

$$\forall X, \nexists f : X \twoheadrightarrow \mathcal{P}(X)$$

□

2.3 Countable and Uncountable Sets

Theorem 2.3.1.

$$\forall X \forall Y, |X| \geq \aleph_0 \land |Y| < \aleph_0 \implies |X \setminus Y| \geq \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $|Y| < \aleph_0$

Assume $|X \setminus Y| < \aleph_0$. By lemma 2.2.15,

$$|(X \setminus Y) \cup Y| < \aleph_0$$

Then

$$\begin{gathered} X = (X \setminus Y) \cup (X \cap Y) \\ \Downarrow \\ X \subseteq (X \setminus Y) \cup Y \\ \Downarrow \qquad \text{Theorem 2.2.5} \\ |X| < \aleph_0 \\ \Downarrow \\ \bot \end{gathered}$$

Therefore,

$$|X \setminus Y| \geq \aleph_0$$

□

Theorem 2.3.2.

$$\forall X \forall Y, |X| \geq \aleph_0 \land \exists f : X \hookrightarrow Y \implies |Y| \geq \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $\exists f : X \hookrightarrow Y$

Assume $|Y| < \aleph_0$. By theorem 2.1.7,

$$f|^{f(X)} : X \xrightarrow{\sim} f(X)$$

Define $h : f(X) \to X$ by

$$h \equiv \left(f|^{f(X)}\right)^{-1}$$

Define $g : Y \to X$ by

$$\forall y \in Y, g(y) := \begin{cases} h(y), & \text{if } y \in f(X) \\ h(y_0), & \text{otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. By theorem 2.1.5,

$$\begin{gathered} h : f(X) \xrightarrow{\sim} X \\ \Downarrow \\ g : Y \twoheadrightarrow X \\ \Downarrow \qquad \text{Theorem 2.2.8} \\ |X| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|Y| \geq \aleph_0$$

$\square$

Lemma 2.3.3.

$$|\mathbb{N}| \geq \aleph_0$$

Proof. Assume $|\mathbb{N}| < \aleph_0$. By theorem 2.2.9,

$$\exists M \in \mathbb{N}, M = \max \mathbb{N}$$

Then

$$\begin{gathered} M < M + 1 \\ \Downarrow \\ M + 1 \notin \mathbb{N} \\ \Downarrow \qquad \text{Axiom 1.1.1} \\ \perp \end{gathered}$$

Therefore,

$$|\mathbb{N}| \geq \aleph_0$$

□

Theorem 2.3.4.

$$\forall X, |X| = \aleph_0 \implies |X| \geq \aleph_0$$

Proof. Suppose $|X| = \aleph_0$. Then

$$\begin{array}{c} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ |X| \geq \aleph_0 \end{array} \qquad \text{Theorem 2.3.2 and lemma 2.3.3}$$

□

Theorem 2.3.5.

$$|\mathbb{Z}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \to \mathbb{Z}$ by

$$\forall n \in \mathbb{N}, f(n) := \begin{cases} -\dfrac{n}{2}, & \text{if } n \equiv 0 \pmod 2 \\ \dfrac{n-1}{2}, & \text{if } n \equiv 1 \pmod 2 \end{cases}$$

Then

- **Injectivity**: Suppose
  - $n_1, n_2 \in \mathbb{N}$
  - $f(n_1) = f(n_2)$

  Then
  - $n \equiv 0 \pmod 2 \implies f(n) = -\dfrac{n}{2} < 0$
  - $n \equiv 1 \pmod 2 \implies f(n) = \dfrac{n-1}{2} \geq 0$

  This implies $n_1 \equiv n_2 \pmod 2$.

- If $n_1 \equiv n_2 \equiv 0 \pmod 2$, then

$$-\frac{n_1}{2} = -\frac{n_2}{2}$$
$$\Downarrow$$
$$n_1 = n_2$$

- If $n_1 \equiv n_2 \equiv 1 \pmod 2$, then

$$\frac{n_1 - 1}{2} = \frac{n_2 - 1}{2}$$
$$\Downarrow$$
$$n_1 = n_2$$

- **Surjectivity**: Suppose $k \in \mathbb{Z}$.
  - If $k \geq 0$, then

$$2k + 1 \in \mathbb{N}$$
$$\Downarrow$$
$$f(2k+1) = \frac{(2k+1) - 1}{2} = \frac{2k}{2} = k$$

  - Otherwise,

$$-2k \in \mathbb{N}$$
$$\Downarrow$$
$$f(-2k) = -\frac{-2k}{2} = k$$

Therefore,

$$f : \mathbb{N} \xrightarrow{\sim} \mathbb{Z}$$
$$\Downarrow$$
$$|\mathbb{Z}| = \aleph_0$$

□

Lemma 2.3.6.

$$\forall X \subseteq \mathbb{N}, |X| \geq \aleph_0 \implies |X| = \aleph_0$$

Proof. Suppose

- $X \subseteq \mathbb{N}$
- $|X| \geq \aleph_0$

Define

- $(X_n)_{n=1}^{\infty}$ in $\mathcal{P}(X)$
- $f : \mathbb{N} \to X$

by

- $X_1 := X$
- $\forall n \in \mathbb{N}, X_{n+1} := X_n \setminus \{f(n)\}$
- $\forall n \in \mathbb{N}, f(n) := \min X_n$

By theorem 2.3.1,

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| \geq \aleph_0 \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists f(n) \in X_n, f(n) = \min X_n \end{gathered} \qquad \text{Theorem 1.1.4}$$

Therefore, f is well-defined.

- **Injectivity**: Suppose
  - $n_1, n_2 \in \mathbb{N}$
  - $n_1 \neq n_2$ (*WLOG* $n_1 < n_2$)

  Then

  $$\begin{gathered} X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\} \\ \Downarrow \\ f(n_2) \neq f(n_1) \end{gathered}$$

- **Surjectivity**: Suppose $x \in X$. Define

  $$S := \{n \in X \mid n < x\} \subseteq [1, x-1]_{\mathbb{N}}$$

  By lemma 2.2.4,

  $$\begin{gathered} |S| < \aleph_0 \\ \Downarrow \\ \exists k \in [0, \infty)_{\mathbb{N}}, |S| = k \end{gathered}$$

  - If $k = 0$, then
    * $S = \varnothing$
    * $x = \min X_1 = f(1)$
  - Otherwise, let
    * $S = \{s_1, s_2, \ldots, s_k\}$
    * $s_1 < s_2 < \cdots < s_k$

Then

$$\begin{array}{cc} \min X_1 = \min S & X_2 = X \setminus \{s_1\} \\ \min X_2 = \min S \setminus \{s_1\} & X_3 = X \setminus \{s_1, s_2\} \\ \vdots & \vdots \\ \min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} & X_{k+1} = X \setminus \{s_1, s_2, \dots, s_k\} = X \setminus S \end{array}$$

Therefore,

$$\begin{gathered} \forall n \in X_{k+1}, n \geq x \\ \Downarrow \\ x = \min X_{k+1} \\ \Downarrow \\ f(k+1) = x \end{gathered}$$

Therefore,

$$\begin{gathered} f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ |X| = \aleph_0 \end{gathered}$$

□

Theorem 2.3.7.

$$\forall X \forall Y, Y \subseteq X \wedge |Y| \geq \aleph_0 \wedge |X| = \aleph_0 \implies |Y| = \aleph_0$$

Proof. Suppose

- $Y \subseteq X$
- $|X| = \aleph_0$
- $|Y| \geq \aleph_0$

Then

$$\begin{gathered} \exists f : \mathbb{N} \xrightarrow{\sim} X \\ \Downarrow \\ f|_{f^{-1}(Y)}^{Y} : f^{-1}(Y) \xrightarrow{\sim} Y \end{gathered} \qquad \text{Theorem 2.1.7}$$

Assume $\left|f^{-1}(Y)\right| < \aleph_0$. By theorem 2.2.1,

$$\begin{gathered} |Y| < \aleph_0 \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\left|f^{-1}(Y)\right| \geq \aleph_0$$

By lemma 2.3.6,

$$\begin{gathered}\left|f^{-1}(Y)\right| = \aleph_0 \\ \Downarrow \\ |Y| = \aleph_0\end{gathered} \qquad \text{Theorem 2.2.1}$$

□

Corollary 2.3.8.

$$\forall X \forall Y, Y \subseteq X \wedge |Y| > \aleph_0 \implies |X| > \aleph_0$$

Proof. Suppose

- $Y \subseteq X$
- $|Y| > \aleph_0$

By corollary 2.2.6,

$$|X| \geq \aleph_0$$

Suppose $|X| = \aleph_0$. By theorem 2.3.7,

$$\begin{gathered}|Y| = \aleph_0 \\ \Downarrow \\ \bot\end{gathered}$$

Therefore,

$$|X| > \aleph_0$$

□

Theorem 2.3.9.

$$\forall X, |X| \geq \aleph_0 \wedge (\exists f : \mathbb{N} \twoheadrightarrow X) \implies |X| = \aleph_0$$

Proof. Suppose

- $|X| \geq \aleph_0$
- $\exists f : \mathbb{N} \twoheadrightarrow X$

Define

$$\forall k \in X, S_k \coloneqq f^{-1}(\{k\}) \neq \varnothing$$

By theorem 1.1.4,

$$\forall k \in X, \exists m_k \in S_k, m_k = \min S_k$$

Define

$$T \coloneqq \{m_k \mid k \in X\} \subseteq \mathbb{N}$$

Define $g : T \to X$ by

$$g \equiv f|_T^X$$

Then

- **Injectivity**: Suppose
  - $m_{k_1}, m_{k_2} \in T$
  - $g(m_{k_1}) = g(m_{k_2})$

  Then

$$\begin{gathered} f(m_{k_1}) = f(m_{k_2}) \\ \Downarrow \\ k_1 = k_2 \\ \Downarrow \\ S_{k_1} = S_{k_2} \\ \Downarrow \\ m_{k_1} = m_{k_2} \end{gathered}$$

- **Surjectivity**: Suppose $k \in X$. Then

$$g(m_k) = k$$

Therefore,

$$\begin{aligned} & g : T \xrightarrow{\sim} X \\ & \Downarrow && \text{Theorems 2.2.1 and 2.3.2} \\ & |T| \geq \aleph_0 \\ & \Downarrow && \text{Lemma 2.3.6} \\ & |T| = \aleph_0 \\ & \Downarrow && \text{Theorem 2.2.1} \\ & |X| = \aleph_0 \end{aligned}$$

□

Theorem 2.3.10.

$$\forall \{X_n\}_{n=1}^{\infty}, (\forall n \in \mathbb{N}, |X_n| < \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| \leq \aleph_0$$

Proof. Suppose

$$\forall n \in \mathbb{N}, |X_n| < \aleph_0$$

Then

$$\forall n \in \mathbb{N}, \exists m_n \in [0, \infty)_{\mathbb{N}}, \exists g_n : [1, m_n]_{\mathbb{N}} \xrightarrow{\sim} X_n$$

Suppose $\left| \bigcup_{n=1}^{\infty} X_n \right| \geq \aleph_0$. Define

- $s_0 := 0$
- $\forall n \in \mathbb{N}, s_n := \sum_{k=1}^{n} m_k$

Then

$$\forall n, m \in \mathbb{N}, n \leq m \implies s_n \leq s_m$$

Assume

$$\exists N \in \mathbb{N}, \forall n \in \mathbb{N}, s_n \leq N$$

Then

$$\begin{array}{cr}
\{s_n\}_{n=1}^{\infty} \subseteq [0, N]_{\mathbb{N}} & \\
\Downarrow & \text{Theorem 1.4.5} \\
\exists n_0 \in \mathbb{N}, s_{n_0} = \max\{s_n\}_{n=1}^{\infty} & \\
\Downarrow & \\
\forall n \in [n_0 + 1, \infty)_{\mathbb{N}}, m_n = 0 & \\
\Downarrow & \\
\bigcup_{n=1}^{\infty} X_n = \bigcup_{n=1}^{n_0} X_n & \\
\Downarrow & \text{Theorem 2.2.16} \\
\left| \bigcup_{n=1}^{\infty} X_n \right| < \aleph_0 & \\
\Downarrow & \\
\perp &
\end{array}$$

Therefore,

$$\forall N \in \mathbb{N}, \exists n \in \mathbb{N}, s_n > N$$

First, we will show

$$\mathbb{N} = \bigcup_{n=1}^{\infty} [s_{n-1} + 1, s_n]_{\mathbb{N}}$$

- ($\subseteq$) Suppose $k \in \mathbb{N}$. Define

$$S_k := \{n \in \mathbb{N} \mid s_n > k\} \neq \varnothing$$

By theorem 1.1.4,

$$\exists n_k \in S_k, n_k = \min S_k$$

Then

$$\begin{gathered} s_{n_k-1} \leq k < s_{n_k} \\ \Downarrow \\ k \in [s_{n_k-1}+1, s_{n_k}]_{\mathbb{N}} \end{gathered}$$

- ($\supseteq$) Suppose $k \in \displaystyle\bigcup_{n=1}^{\infty} [s_{n-1}+1, s_n]_{\mathbb{N}}$. Then

$$\begin{gathered} \exists n \in \mathbb{N}, k \in [s_{n-1}+1, s_n]_{\mathbb{N}} \\ \Downarrow \\ k \in \mathbb{N} \end{gathered}$$

Next, we will show

$$\forall i, j \in \mathbb{N}, i \neq j \implies [s_{i-1}+1, s_i]_{\mathbb{N}} \cap [s_{j-1}+1, s_j]_{\mathbb{N}} = \varnothing$$

Suppose

- $i, j \in \mathbb{N}$
- $i \neq j$ (*WLOG* $i < j$)

Then

$$\begin{gathered} s_i \leq s_{j-1} \\ \Downarrow \\ [s_{i-1}+1, s_i]_{\mathbb{N}} \cap [s_{j-1}+1, s_j]_{\mathbb{N}} = \varnothing \end{gathered}$$

Therefore,

$$\forall k \in \mathbb{N}, \exists! n \in \mathbb{N}, \exists! m \in [1, m_n]_{\mathbb{N}}, k = s_{n-1} + m$$

Define $h : \mathbb{N} \to \displaystyle\bigcup_{n=1}^{\infty} X_n$ by

$$\forall n \in \mathbb{N}, \forall m \in [1, m_n]_{\mathbb{N}}, h(s_{n-1}+m) := g_n(m)$$

Suppose $x \in \displaystyle\bigcup_{n=1}^{\infty} X_n$. Then

$$\begin{gathered} \exists n \in \mathbb{N}, x \in X_n \\ \Downarrow \\ \exists m \in [1, m_n]_{\mathbb{N}}, g_n(m) = x \\ \Downarrow \\ h(s_{n-1}+m) = x \end{gathered} \qquad g_n : [1, m_n]_{\mathbb{N}} \xrightarrow{\sim} X_n$$

Since $x \in \bigcup_{n=1}^{\infty} X_n$ was arbitrary,

$$\begin{array}{c} h : \mathbb{N} \twoheadrightarrow \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{array} \qquad \text{Theorem 2.3.9}$$

□

Remark. Strictly speaking, theorem 2.3.10 requires the **axiom of choice** to prove. In this textbook, the **axiom of choice** is not discussed, so we will just assume that the proof is valid. Similar issues arise for theorems 2.3.11 and 2.3.14.

Theorem 2.3.11.

$$\forall \{X_n\}_{n=1}^{\infty}, (\forall n \in \mathbb{N}, |X_n| = \aleph_0) \implies \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0$$

Proof. Suppose

$$\forall n \in \mathbb{N}, |X_n| = \aleph_0$$

Then

$$\forall n \in \mathbb{N}, \exists f_n : \mathbb{N} \xrightarrow{\sim} X_n$$

Assume $\left| \bigcup_{n=1}^{\infty} X_n \right| < \aleph_0$. Then

$$\begin{array}{cr} X_1 \subseteq \bigcup_{n=1}^{\infty} X_n & \\ \Downarrow & \text{Theorem 2.2.5} \\ |X_1| < \aleph_0 & \\ \Downarrow & \text{Theorem 2.3.4} \\ \perp & \end{array}$$

Therefore,

$$\left| \bigcup_{n=1}^{\infty} X_n \right| \geq \aleph_0$$

Define

$$\forall n \in \mathbb{N}, T_n := \left\{ k \in \mathbb{N} \,\middle|\, n \leq \frac{k(k+1)}{2} \right\} \subseteq \mathbb{N}$$

Then

$$\begin{gathered}\forall n \in \mathbb{N}, 2n \in T_n \\ \Downarrow \\ \forall n \in \mathbb{N}, \exists t_n \in T_n, t_n = \min T_n\end{gathered} \qquad \text{Theorem 1.1.4}$$

Define

$$\forall n \in \mathbb{N}, s_n := n - \frac{t_n(t_n - 1)}{2}$$

Then

$$\begin{gathered}t_n - 1 \notin T_n \\ \Downarrow \\ n > \frac{t_n(t_n - 1)}{2} \\ \Downarrow \\ s_n \in \mathbb{N}\end{gathered}$$

Also,

$$\begin{gathered}t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \\ \Downarrow \\ t_n - s_n + 1 \geq 1 \\ \Downarrow \\ t_n - s_n + 1 \in \mathbb{N}\end{gathered} \qquad t_n \in T_n$$

Define $g : \mathbb{N} \to \displaystyle\bigcup_{n=1}^{\infty} X_n$ by

$$\forall n \in \mathbb{N}, g(n) := f_{s_n}(t_n - s_n + 1)$$

Suppose $x \in \displaystyle\bigcup_{n=1}^{\infty} X_n$. Then

$$\begin{gathered}\exists m \in \mathbb{N}, x \in X_m \\ \Downarrow \\ \exists l \in \mathbb{N}, f_m(l) = x\end{gathered} \qquad f_m : \mathbb{N} \xrightarrow{\sim} X_m$$

Define

$$k := \frac{(l + m - 1)(l + m - 2)}{2} + m \in \mathbb{N}$$

Then

- $k - \dfrac{(l + m - 2)(l + m - 1)}{2} = m > 0$
- $k - \dfrac{(l + m - 1)(l + m)}{2} = 1 - l \leq 0$

This implies

$$\begin{gathered} l+m-2 \notin T_k \wedge l+m-1 \in T_k \\ \Downarrow \\ t_k = l+m-1 \\ \Downarrow \\ s_k = m \\ \Downarrow \\ g(k) = f_m(l) = x \end{gathered}$$

Since $x \in \bigcup_{n=1}^{\infty} X_n$ was arbitrary,

$$\begin{gathered} g : \mathbb{N} \twoheadrightarrow \bigcup_{n=1}^{\infty} X_n \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.3.9}$$

□

Corollary 2.3.12.

$$\forall n \in \mathbb{N}, \forall \{X_k\}_{k=1}^{n}, (\forall k \in [1,n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left| \bigcup_{k=1}^{n} X_k \right| = \aleph_0$$

Proof. Suppose

$$\forall k \in [1,n]_{\mathbb{N}}, |X_k| = \aleph_0$$

Define

$$\forall k \in [n+1,\infty)_{\mathbb{N}}, X_k := X_n$$

Then

$$\begin{gathered} \forall n \in \mathbb{N}, |X_n| = \aleph_0 \\ \Downarrow \\ \left| \bigcup_{n=1}^{\infty} X_n \right| = \aleph_0 \\ \Downarrow \\ \left| \bigcup_{k=1}^{n} X_k \right| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.3.11}$$

□

Lemma 2.3.13.

$$\forall X \forall Y, |X| < \aleph_0 \wedge |Y| = \aleph_0 \implies |X \cup Y| = \aleph_0$$

Proof. Suppose

- $|X| < \aleph_0$
- $|Y| = \aleph_0$

Then

- $\exists f : \mathbb{N} \xrightarrow{\sim} Y$
- $\exists m \in [0, \infty)_{\mathbb{N}}, \exists g : [1, m]_{\mathbb{N}} \xrightarrow{\sim} X$

Also,

$$\begin{array}{c} Y \subseteq X \cup Y \\ \Downarrow \\ |X \cup Y| \geq \aleph_0 \end{array}$$ Corollary 2.2.6 and theorem 2.3.4

Define $h : \mathbb{N} \to X \cup Y$ by

- $\forall n \in [1, m]_{\mathbb{N}}, h(n) := g(n)$
- $\forall n \in [m+1, \infty)_{\mathbb{N}}, h(n) := f(n - m)$

Suppose $x \in X \cup Y$. Then

- If $x \in X$, then
$$\exists n \in [1, m]_{\mathbb{N}}, g(n) = x$$
This implies
$$h(n) = x$$
- Otherwise,
$$\exists n \in \mathbb{N}, f(n) = x$$
This implies
$$h(n + m) = x$$

Since $x \in X \cup Y$ was arbitrary,

$$\begin{array}{c} h : \mathbb{N} \twoheadrightarrow X \cup Y \\ \Downarrow \\ |X \cup Y| = \aleph_0 \end{array}$$ Theorem 2.3.9

$\square$

Theorem 2.3.14.

$$\forall Y, \forall (X_\alpha)_{\alpha \in Y}, |Y| \leq \aleph_0 \wedge (\forall \alpha \in Y, |X_\alpha| \leq \aleph_0) \implies \left| \bigcup_{\alpha \in Y} X_\alpha \right| \leq \aleph_0$$

Proof. Suppose

- $|Y| \leq \aleph_0$
- $\forall \alpha \in Y, |X_\alpha| \leq \aleph_0$

Then

- If
$$\forall \alpha \in Y, |X_\alpha| < \aleph_0$$
then
  - If $|Y| < \aleph_0$, then
  $$\exists n \in [0, \infty)_{\mathbb{N}}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} Y$$
  This implies
  $$\begin{gathered} \bigcup_{\alpha \in Y} X_\alpha = \bigcup_{k=1}^{n} X_{f(k)} \\ \Downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| < \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.16}$$
  - Otherwise,
  $$\exists g : \mathbb{N} \xrightarrow{\sim} Y$$
  This implies
  $$\begin{gathered} \bigcup_{\alpha \in Y} X_\alpha = \bigcup_{n=1}^{\infty} X_{g(n)} \\ \Downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| \leq \aleph_0 \end{gathered} \qquad \text{Theorem 2.3.10}$$
- Otherwise,
$$\exists \beta \in Y, |X_\beta| = \aleph_0$$
Define
$$\forall \alpha \in Y, Y_\alpha \coloneqq X_\alpha \cup X_\beta$$

By corollary 2.3.12 and lemma 2.3.13,

$$\forall \alpha \in Y, |Y_\alpha| = \aleph_0$$

Then

- If $|Y| < \aleph_0$, then

$$\exists n \in \mathbb{N}, \exists f : [1, n]_{\mathbb{N}} \xrightarrow{\sim} Y$$

This implies

$$\begin{aligned} \bigcup_{\alpha \in Y} X_\alpha = \bigcup_{\alpha \in Y} Y_\alpha = \bigcup_{k=1}^{n} Y_{f(k)} \\ \Downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| = \aleph_0 \end{aligned} \qquad \text{Corollary 2.3.12}$$

- Otherwise,

$$\exists g : \mathbb{N} \xrightarrow{\sim} Y$$

This implies

$$\begin{aligned} \bigcup_{\alpha \in Y} X_\alpha = \bigcup_{\alpha \in Y} Y_\alpha = \bigcup_{n=1}^{\infty} Y_{g(n)} \\ \Downarrow \\ \left| \bigcup_{\alpha \in Y} X_\alpha \right| = \aleph_0 \end{aligned} \qquad \text{Theorem 2.3.11}$$

□

Lemma 2.3.15.

$$|\mathbb{N} \times \mathbb{N}| = \aleph_0$$

Proof. Define $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by

$$\forall m, n \in \mathbb{N}, f(m, n) := \frac{(m+n-1)(m+n-2)}{2} + m$$

Then

- **Injectivity**: Suppose

- $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$
- $(m_1, n_1) \neq (m_2, n_2)$

Then

- If $m_1 + n_1 \neq m_2 + n_2$ (*WLOG* $m_1 + n_1 < m_2 + n_2$), then

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2) \end{aligned}$$

- Otherwise,

$$\begin{gathered} m_1 \neq m_2 \\ \Downarrow \\ f(m_1, n_1) \neq f(m_2, n_2) \end{gathered}$$

- **Surjectivity**: Suppose $k \in \mathbb{N}$. Define t_k, s_k as theorem 2.3.11. Then

$$f(s_k, t_k - s_k + 1) = k$$

Therefore,

$$\begin{gathered} f : \mathbb{N} \times \mathbb{N} \xrightarrow{\sim} \mathbb{N} \\ \Downarrow \\ |\mathbb{N} \times \mathbb{N}| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.1}$$

$\square$

Theorem 2.3.16.

$$\forall X \forall Y, |X| = \aleph_0 \wedge |Y| = \aleph_0 \implies |X \times Y| = \aleph_0$$

Proof. Suppose

- $|X| = \aleph_0$
- $|Y| = \aleph_0$

Then

- $\exists f : \mathbb{N} \xrightarrow{\sim} X$
- $\exists g : \mathbb{N} \xrightarrow{\sim} Y$

Define $h : \mathbb{N} \times \mathbb{N} \to X \times Y$ by

$$\forall m, n \in \mathbb{N}, h(m, n) := (f(m), g(n))$$

Then

- **Injectivity**: Suppose
  - $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$
  - $h(m_1, n_1) = h(m_2, n_2)$

  Then
$$\begin{gathered} f(m_1) = f(m_2) \wedge g(n_1) = g(n_2) \\ \Downarrow \\ m_1 = m_2 \wedge n_1 = n_2 \\ \Downarrow \\ (m_1, n_1) = (m_2, n_2) \end{gathered}$$

- **Surjectivity**: Suppose $(x, y) \in X \times Y$. Then
$$\begin{gathered} \exists m, n \in \mathbb{N}, f(m) = x \wedge g(n) = y \\ \Downarrow \\ h(m, n) = (x, y) \end{gathered}$$

Therefore,
$$\begin{gathered} h : \mathbb{N} \times \mathbb{N} \xrightarrow{\sim} X \times Y \\ \Downarrow \qquad \text{Theorem 2.2.1 and lemma 2.3.15} \\ |X \times Y| = \aleph_0 \end{gathered}$$

$\square$

Theorem 2.3.17.

$$\forall n \in \mathbb{N}, \forall \{X_k\}_{k=1}^{n}, (\forall k \in [1, n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left| \prod_{k=1}^{n} X_k \right| = \aleph_0$$

Proof. Define

$$\forall n \in \mathbb{N}, P(n) \iff \forall \{X_k\}_{k=1}^{n}, (\forall k \in [1,n]_{\mathbb{N}}, |X_k| = \aleph_0) \implies \left|\prod_{k=1}^{n} X_k\right| = \aleph_0$$

Then

- **Base case:** Suppose $|X_1| = \aleph_0$. Then

$$\begin{gathered} X_1 = \prod_{k=1}^{1} X_k \\ \Downarrow \\ \left|\prod_{k=1}^{1} X_k\right| = \aleph_0 \\ \Downarrow \\ P(1) \end{gathered}$$

- **Inductive step:** Suppose
  - $P(n)$
  - $\forall k \in [1, n+1]_{\mathbb{N}}, |X_k| = \aleph_0$

  Since $P(n)$,

$$\left|\prod_{k=1}^{n} X_k\right| = \aleph_0$$

  By lemma 2.2.18,

$$\begin{gathered} \prod_{k=1}^{n+1} X_k \sim \left(\prod_{k=1}^{n} X_k\right) \times X_{n+1} \\ \Downarrow \quad \text{Theorem 2.3.16} \\ \left|\prod_{k=1}^{n+1} X_k\right| = \aleph_0 \\ \Downarrow \\ P(n+1) \end{gathered}$$

By theorem 1.1.2,

$$\forall n \in \mathbb{N}, P(n)$$

□

Theorem 2.3.18.
$$|\mathbb{Q}| = \aleph_0$$

Proof. Define $f : \mathbb{Z} \times \mathbb{N} \to \mathbb{Q}$ by

$$\forall (m, n) \in \mathbb{Z} \times \mathbb{N}, f(m, n) := \frac{m}{n}$$

Then
$$f : \mathbb{Z} \times \mathbb{N} \twoheadrightarrow \mathbb{Q}$$

By theorems 2.3.5 and 2.3.16,

$$\begin{gathered} |\mathbb{Z} \times \mathbb{N}| = \aleph_0 \\ \Downarrow \\ \exists g : \mathbb{N} \xrightarrow{\sim} \mathbb{Z} \times \mathbb{N} \end{gathered}$$

Also,
$$\begin{gathered} \mathbb{N} \subseteq \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| \geq \aleph_0 \end{gathered} \qquad \text{Corollary 2.2.6 and lemma 2.3.3}$$

By theorem 2.1.9,
$$\begin{gathered} f \circ g : \mathbb{N} \twoheadrightarrow \mathbb{Q} \\ \Downarrow \\ |\mathbb{Q}| = \aleph_0 \end{gathered} \qquad \text{Theorem 2.3.9}$$

□

Lemma 2.3.19.
$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

Proof. Define

$$\forall n, m \in \mathbb{N}, e_{nm} := \begin{cases} 1, & \text{if } m = n \\ 0, & \text{otherwise} \end{cases}$$

Define $f : \mathbb{N} \to \prod_{n=1}^{\infty} \{0, 1\}$ by

$$\forall n \in \mathbb{N}, f(n) := (e_{nm})_{m=1}^{\infty}$$

Suppose

- $n_1, n_2 \in \mathbb{N}$
- $f(n_1) = f(n_2)$

Then

$$\begin{gathered} \forall m \in \mathbb{N}, e_{n_1 m} = e_{n_2 m} \\ \Downarrow \\ e_{n_1 n_1} = e_{n_2 n_1} = 1 \\ \Downarrow \\ n_2 = n_1 \end{gathered}$$

Therefore,

$$f : \mathbb{N} \hookrightarrow \prod_{n=1}^{\infty} \{0, 1\}$$

By theorem 2.3.2 and lemma 2.3.3,

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| \geq \aleph_0$$

Assume $\left| \prod_{n=1}^{\infty} \{0, 1\} \right| = \aleph_0$. Then

$$\exists f : \mathbb{N} \xrightarrow{\sim} \prod_{n=1}^{\infty} \{0, 1\}$$

Define

- $\forall n \in \mathbb{N}, f(n) = (a_{nm})_{m=1}^{\infty}$
- $\forall m \in \mathbb{N}, b_m := 1 - a_{mm}$

Then

$$(b_m)_{m=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\}$$

Also,

$$\begin{gathered} \forall n \in \mathbb{N}, (b_m)_{m=1}^{\infty} \neq f(n) \\ \Downarrow \qquad\qquad f : \mathbb{N} \xrightarrow{\sim} \prod_{n=1}^{\infty} \{0, 1\} \\ \bot \end{gathered}$$

Therefore,

$$\left| \prod_{n=1}^{\infty} \{0, 1\} \right| > \aleph_0$$

$\square$

Theorem 2.3.20.
$$|\mathbb{R}| > \aleph_0$$

Proof. Define $f : \mathbb{N} \to [0,1)$ by
$$\forall n \in \mathbb{N}, f(n) := \frac{1}{n+1}$$

Suppose

- $n_1, n_2 \in \mathbb{N}$
- $f(n_1) = f(n_2)$

Then
$$\frac{1}{n_1+1} = \frac{1}{n_2+1}$$
$$\Downarrow$$
$$n_1 = n_2$$

Therefore,
$$f : \mathbb{N} \hookrightarrow [0,1)$$

By lemma 2.3.3 and theorem 2.3.2,
$$|[0,1)| \geq \aleph_0$$

Define $g : [0,1) \to \prod_{n=1}^{\infty} \{0,1\}$ by
$$\forall x \in [0,1), g(x) := \mathcal{G}_2(x)$$

By theorem 1.6.13, g is well-defined. By theorem 1.6.14,
$$g : [0,1) \hookrightarrow \prod_{n=1}^{\infty} \{0,1\}$$

Define
$$Y := g([0,1)) \subseteq \prod_{n=1}^{\infty} \{0,1\}$$

By theorem 2.1.7,
$$g|^{Y} : [0,1) \xrightarrow{\sim} Y$$

Assume $|[0,1)| = \aleph_0$. By theorem 2.2.1,
$$|Y| = \aleph_0$$

Define

$$Z := \prod_{n=1}^{\infty} \{0,1\} \setminus Y$$

By theorems 1.6.15 and 1.6.16,

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \;\middle|\; \exists N \in \mathbb{N}, \forall n \in [N,\infty)_{\mathbb{N}}, a_n = 1 \right\}$$

Define

$$\forall N \in \mathbb{N}, Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0,1\} \;\middle|\; \forall n \in [N,\infty)_{\mathbb{N}}, a_n = 1 \right\} \neq \varnothing$$

Define

$$\forall N \in \mathbb{N}, F_N : \prod_{n=1}^{N-1} \{0,1\} \to Z_N$$

by

$$\forall N \in \mathbb{N}, \forall (a_1, \ldots, a_{N-1}) \in \prod_{n=1}^{N-1} \{0,1\},$$

$$F_N(a_1, \ldots, a_{N-1}) = (a_1, \ldots, a_{N-1}, 1, 1, 1, \ldots)$$

Suppose

- $N \in \mathbb{N}$
- $(a_1, \ldots, a_{N-1}, 1, 1, 1, \ldots) \in Z_N$

Then

$$F_N(a_1, \ldots, a_{N-1}) = (a_1, \ldots, a_{N-1}, 1, 1, 1, \ldots)$$

Therefore,

$$\forall N \in \mathbb{N}, F_N : \prod_{n=1}^{N-1} \{0,1\} \twoheadrightarrow Z_N$$

By theorem 2.2.19,

$$\forall N \in \mathbb{N}, \left| \prod_{n=1}^{N-1} \{0,1\} \right| < \aleph_0$$

$$\Downarrow \qquad \text{Theorem 2.2.8}$$

$$\forall N \in \mathbb{N}, |Z_N| < \aleph_0$$

This implies

$$Z = \bigcup_{N=1}^{\infty} Z_N$$

$$\Downarrow \qquad \text{Theorem 2.3.10}$$

$$|Z| \leq \aleph_0$$

Then

$$\begin{array}{cr} \prod_{n=1}^{\infty}\{0,1\} = Y \cup Z & \\ \Downarrow & \text{Theorem 2.3.14} \\ \left|\prod_{n=1}^{\infty}\{0,1\}\right| \leq \aleph_0 & \\ \Downarrow & \text{Lemma 2.3.19} \\ \perp & \end{array}$$

Therefore,

$$\begin{array}{cr} |[0,1)| > \aleph_0 & \\ \Downarrow & \text{Corollary 2.3.8} \\ |\mathbb{R}| > \aleph_0 & \end{array}$$

□

2.4 Metric Spaces

Definition 2.4.1. A function $d : X \times X \to [0, \infty)$ is called a **metric** on X if

(A) $\forall p, q \in X, d(p, q) = 0 \iff p = q$

(B) $\forall p, q \in X, d(p, q) = d(q, p)$

(C) $\forall p, q, r \in X, d(p, r) \leq d(p, q) + d(q, r)$

(C) is called the **triangle inequality**. The structure (X, d) is called a **metric space**.

Theorem 2.4.2. *Suppose $n \in \mathbb{N}$. Define $d : \mathbb{R}^n \times \mathbb{R}^n \to [0, \infty)$ by*

$$\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n, d(\boldsymbol{x}, \boldsymbol{y}) := \|\boldsymbol{x} - \boldsymbol{y}\|$$

Then d is a metric on $\mathbb{R}^n$.

Proof. By definition 2.4.1,

- Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. By definition 1.9.3,

$$\|\boldsymbol{x} - \boldsymbol{y}\| = 0 \iff \forall k \in [1, n]_{\mathbb{N}}, x_k = y_k \iff \boldsymbol{x} = \boldsymbol{y}$$

- Suppose $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. By definition 1.9.3,

$$\|\boldsymbol{x}-\boldsymbol{y}\| = \sqrt{\sum_{k=1}^{n}(x_k - y_k)^2} = \sqrt{\sum_{k=1}^{n}(y_k - x_k)^2} = \|\boldsymbol{y}-\boldsymbol{x}\|$$

- Suppose $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n$. By theorem 1.9.4,

$$\|\boldsymbol{x}-\boldsymbol{z}\| = \|(\boldsymbol{x}-\boldsymbol{y}) + (\boldsymbol{y}-\boldsymbol{z})\| \leq \|\boldsymbol{x}-\boldsymbol{y}\| + \|\boldsymbol{y}-\boldsymbol{z}\|$$

□

Remark. Throughout this textbook, **euclidean spaces** use the **standard metric** defined in theorem 2.4.2, unless otherwise specified.

Definition 2.4.3. Suppose

- $p \in |(X, d)|$
- $r \in \mathbb{R}^+$

Then

- The **neighborhood** of p with radius r is defined by

$$N_r(p) \coloneqq \{q \in X \mid d(p,q) < r\}$$

- The **deleted neighborhood** of p with radius r is defined by

$$\widetilde{N}_r(p) \coloneqq N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p,q) < r\}$$

Definition 2.4.4. Suppose

- $E \subseteq |(X, d)|$
- $p \in X$

Then

- p is called an **limit point** of E if

$$\forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing$$

  The set of all limit points of E is denoted by

$$E' \coloneqq \left\{p \in X \,\middle|\, \forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing\right\}$$

- p is called an **interior point** of E if

  $$\exists r \in \mathbb{R}^+, N_r(p) \subseteq E$$

  The set of all interior points of E is denoted by

  $$E^\circ := \left\{p \in X \;\middle|\; \exists r \in \mathbb{R}^+, N_r(p) \subseteq E\right\}$$

- p is called a **isolated point** of E if

  $$\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}$$

  The set of all isolated points of E is denoted by

  $$E^i := \left\{p \in X \;\middle|\; \exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\}\right\}$$

Definition 2.4.5. Suppose $E \subseteq |(X,d)|$. Then

- E is said to be **open** if

  $$\forall p \in E, p \in E^\circ$$

  The set of all open sets in X is denoted by

  $$\mathcal{O}(X) := \{E \subseteq X \mid \forall p \in E, p \in E^\circ\}$$

- E is said to be **closed** if

  $$E' \subseteq E$$

  The set of all closed sets in X is denoted by

  $$\mathcal{C}(X) := \left\{E \subseteq X \;\middle|\; E' \subseteq E\right\}$$

Definition 2.4.6. Suppose $E \subseteq |(X,d)|$. The **closure** $\overline{E}$ of E is defined by

$$\overline{E} := E \cup E'$$

Definition 2.4.7. Suppose $E \subseteq |(X,d)|$. Then

- E is said to be **bounded** if

  $$X \neq \varnothing \implies \exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)$$

  The set of all bounded sets in X is denoted by

  $$\mathcal{B}(X) := \left\{E \subseteq X \;\middle|\; X \neq \varnothing \implies \exists p \in X, \exists r \in \mathbb{R}^+, E \subseteq N_r(p)\right\}$$

- E is said to be **perfect** if
$$E = E'$$

- E is said to be **dense** if
$$\overline{E} = X$$

Theorem 2.4.8.
$$\forall p \in |(X,d)|, \forall r \in \mathbb{R}^+, N_r(p) \in \mathcal{O}(X)$$

Proof. Suppose $q \in N_r(p)$. Then

$$\begin{gathered} d(p,q) < r \\ \Downarrow \\ \varepsilon := r - d(p,q) > 0 \end{gathered}$$

Suppose $s \in N_\varepsilon(q)$. Then

$$\begin{gathered} d(q,s) < \varepsilon \\ \Downarrow \\ d(p,s) \leq d(p,q) + d(q,s) < d(p,q) + \varepsilon = r \\ \Downarrow \\ s \in N_r(p) \end{gathered}$$

Since $s \in N_\varepsilon(q)$ was arbitrary,

$$\begin{gathered} N_\varepsilon(q) \subseteq N_r(p) \\ \Downarrow \\ q \in (N_r(p))^\circ \end{gathered}$$

□

Theorem 2.4.9.

$$\forall E \subseteq |(X,d)|, \forall p \in E', \forall r \in \mathbb{R}^+, |E \cap N_r(p)| \geq \aleph_0$$

Proof. Suppose $E \subseteq X$. Assume

$$\exists p \in E', \exists r \in \mathbb{R}^+, |E \cap N_r(p)| < \aleph_0$$

Define

$$S := E \cap \widetilde{N}_r(p) \neq \varnothing$$

Then

$$\begin{gathered} S \subseteq E \cap N_r(p) \\ \Downarrow \\ |S| < \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.5}$$

Define

$$T := \{d(p, q) \mid q \in S\} \neq \varnothing$$

Then

$$\begin{gathered} p \notin S \\ \Downarrow \\ T \subseteq \mathbb{R}^+ \end{gathered}$$

Define $f : S \to T$ by

$$\forall q \in S, f(q) := d(p, q)$$

Then

$$\begin{gathered} \forall t \in T, \exists q \in S, t = d(p, q) \\ \Downarrow \\ f : S \twoheadrightarrow T \\ \Downarrow \qquad \text{Theorem 2.2.8} \\ |T| < \aleph_0 \\ \Downarrow \qquad \text{Theorem 2.2.9} \\ \exists \varepsilon \in \mathbb{R}^+, \varepsilon = \min T \end{gathered}$$

This implies

$$\begin{gathered} \forall q \in S, d(p, q) \geq \varepsilon \\ \Downarrow \\ E \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$|E \cap N_r(p)| \geq \aleph_0$$

□

Theorem 2.4.10.

$$\forall E \subseteq |(X, d)|, E \in \mathcal{O}(X) \iff E^c \in \mathcal{C}(X)$$

Proof. ($\Longrightarrow$) Suppose

- $E \in \mathcal{O}(X)$
- $p \in (E^c)'$

Assume $p \in E$. Then

$$\begin{gathered}\exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(p) \subseteq E \\ \Downarrow \\ E^c \cap \widetilde{N}_\varepsilon(p) = \varnothing \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$p \in E^c$$

Since $p \in (E^c)'$ was arbitrary,

$$\begin{gathered}(E^c)' \subseteq E^c \\ \Downarrow \\ E^c \in \mathcal{C}(X)\end{gathered}$$

($\Longleftarrow$) Suppose

- $E^c \in \mathcal{C}(X)$
- $p \in E$

Assume

$$\nexists r \in \mathbb{R}^+, N_r(p) \subseteq E,$$

Then

$$\begin{gathered}p \notin E^c \\ \Downarrow \\ \forall r \in \mathbb{R}^+, E^c \cap \widetilde{N}_r(p) \neq \varnothing \\ \Downarrow \\ p \in (E^c)' \\ \Downarrow \qquad E^c \in \mathcal{C}(X) \\ p \in E^c \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\begin{gathered}\exists r \in \mathbb{R}^+, N_r(p) \subseteq E \\ \Downarrow \\ p \in E^\circ\end{gathered}$$

$\square$

Theorem 2.4.11.
$$\forall \{E_y\}_{y \in Y} \subseteq \mathcal{O}(|(X,d)|), \bigcup_{y \in Y} E_y \in \mathcal{O}(X)$$

Proof. Suppose

- $\{E_y\}_{y \in Y} \subseteq \mathcal{O}(X)$
- $p \in \bigcup_{y \in Y} E_y$

Then
$$\begin{gathered} \exists y_0 \in Y, p \in E_{y_0} \\ \Downarrow \\ \exists r \in \mathbb{R}^+, N_r(p) \subseteq E_{y_0} \\ \Downarrow \\ N_r(p) \subseteq \bigcup_{y \in Y} E_y \\ \Downarrow \\ p \in \left(\bigcup_{y \in Y} E_y\right)^{\circ} \end{gathered}$$

□

Corollary 2.4.12.
$$\forall \{E_y\}_{y \in Y} \subseteq \mathcal{C}(|(X,d)|), \bigcap_{y \in Y} E_y \in \mathcal{C}(X)$$

Proof. Suppose $\{E_y\}_{y \in Y} \subseteq \mathcal{C}(X)$. By theorem 2.4.10,
$$\begin{gathered} \forall y \in Y, (E_y)^c \in \mathcal{O}(X) \\ \Downarrow \quad \text{Theorem 2.4.11} \\ \bigcup_{y \in Y} (E_y)^c \in \mathcal{O}(X) \\ \Downarrow \quad \text{Theorem 2.4.10} \\ \bigcap_{y \in Y} E_y = \left(\bigcup_{y \in Y} (E_y)^c\right)^c \in \mathcal{C}(X) \end{gathered}$$

□

Theorem 2.4.13.

$$\forall n \in \mathbb{N}, \forall \{E_k\}_{k=1}^n \subseteq \mathcal{O}(|(X,d)|), \bigcap_{k=1}^n E_k \in \mathcal{O}(X)$$

Proof. Suppose

- $n \in \mathbb{N}$
- $\{E_k\}_{k=1}^n \subseteq \mathcal{O}(X)$
- $p \in \bigcap_{k=1}^n E_k$

Then

$$\begin{gathered} \forall j \in [1,n]_{\mathbb{N}}, p \in E_j \\ \Downarrow \\ \forall j \in [1,n]_{\mathbb{N}}, \exists r_j \in \mathbb{R}^+, N_{r_j}(p) \subseteq E_j \end{gathered}$$

Define

$$R \coloneqq \{r_j \mid j \in [1,n]_{\mathbb{N}}\} \subseteq \mathbb{R}^+$$

Define $f : [1,n]_{\mathbb{N}} \to R$ by

$$\forall j \in [1,n]_{\mathbb{N}}, f(j) \coloneqq r_j$$

Then

$$\begin{gathered} \forall \varepsilon \in R, \exists j \in [1,n]_{\mathbb{N}}, \varepsilon = r_j \\ \Downarrow \\ f : [1,n]_{\mathbb{N}} \twoheadrightarrow R \\ \Downarrow \qquad \text{Lemma 2.2.4 and theorem 2.2.8} \\ |R| < \aleph_0 \\ \Downarrow \qquad \text{Theorem 2.2.9} \\ \exists r \in \mathbb{R}^+, r = \min R \\ \Downarrow \\ \forall j \in [1,n]_{\mathbb{N}}, N_r(p) \subseteq E_j \\ \Downarrow \\ N_r(p) \subseteq \bigcap_{k=1}^n E_k \\ \Downarrow \\ p \in \left(\bigcap_{k=1}^n E_k\right)^\circ \end{gathered}$$

□

Corollary 2.4.14.

$$\forall n \in \mathbb{N}, \forall \{E_k\}_{k=1}^{n} \subseteq \mathcal{C}(|(X,d)|), \bigcup_{k=1}^{n} E_k \in \mathcal{C}(X)$$

Proof. Suppose

- $n \in \mathbb{N}$
- $\{E_k\}_{k=1}^{n} \subseteq \mathcal{C}(X)$

By theorem 2.4.10,

$$\begin{array}{cr} \forall k \in [1,n]_{\mathbb{N}}, (E_k)^c \in \mathcal{O}(X) & \\ \Downarrow & \text{Theorem 2.4.13} \\ \bigcap_{k=1}^{n} (E_k)^c \in \mathcal{O}(X) & \\ \Downarrow & \text{Theorem 2.4.10} \\ \bigcup_{k=1}^{n} E_k = \left(\bigcap_{k=1}^{n} (E_k)^c\right)^c \in \mathcal{C}(X) & \end{array}$$

□

Theorem 2.4.15. *Suppose $E \subseteq |(X,d)|$. Then*

(A) $\overline{E} \in \mathcal{C}(X)$

(B) $E = \overline{E} \iff E \in \mathcal{C}(X)$

(C) $\forall F \in \mathcal{C}(X), E \subseteq F \implies \overline{E} \subseteq F$

Proof.

(A) Suppose $p \in \left(\overline{E}\right)^c$. Then

$$\begin{array}{cr} p \notin E' & \\ \Downarrow & \\ \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) = \varnothing & \\ \Downarrow & p \notin E \\ E \cap N_r(p) = \varnothing & \\ \Downarrow & \\ N_r(p) \subseteq E^c & \end{array}$$

Suppose $q \in N_r(p)$. By theorem 2.4.8,

$$\begin{gathered}\exists\varepsilon \in \mathbb{R}^+, N_\varepsilon(q) \subseteq N_r(p)\\ \Downarrow\\ E \cap N_\varepsilon(q) = \varnothing\\ \Downarrow\\ E \cap \widetilde{N}_\varepsilon(q) = \varnothing\\ \Downarrow\\ q \notin E'\end{gathered}$$

Since $q \in N_r(p)$ was arbitrary,

$$\begin{gathered}N_r(p) \subseteq \left(E'\right)^c\\ \Downarrow\\ N_r(p) \subseteq \left(\overline{E}\right)^c\\ \Downarrow\\ p \in \left(\left(\overline{E}\right)^c\right)^\circ\end{gathered} \qquad N_r(p) \subseteq E^c$$

Since $p \in \left(\overline{E}\right)^c$ was arbitrary,

$$\begin{gathered}\left(\overline{E}\right)^c \in \mathcal{O}(X)\\ \Downarrow\\ \overline{E} \in \mathcal{C}(X)\end{gathered} \qquad \text{Theorem 2.4.10}$$

(B) $E \in \mathcal{C}(X) \iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$

(C) Suppose

- $F \in \mathcal{C}(X)$
- $E \subseteq F$
- $p \in E'$

Then

$$\begin{gathered}\forall r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) \neq \varnothing\\ \Downarrow\\ \forall r \in \mathbb{R}^+, F \cap \widetilde{N}_r(p) \neq \varnothing\\ \Downarrow\\ p \in F'\end{gathered}$$

Since $p \in E'$ was arbitrary,

$$\begin{gathered}E' \subseteq F'\\ \Downarrow\\ E' \subseteq F\\ \Downarrow\\ \overline{E} \subseteq F\end{gathered} \qquad F \in \mathcal{C}(X)$$

□

Theorem 2.4.16. *Suppose*

- $E \subseteq \mathbb{R}$
- $E \neq \varnothing$
- $E^u \neq \varnothing$

Then

(A) $\exists \alpha \in \overline{E}, \alpha = \sup E$

(B) $E \in \mathcal{C}(\mathbb{R}) \implies \exists \alpha \in E, \alpha = \sup E$

Proof.

(A) By axiom 1.6.1,

$$\exists \alpha \in \mathbb{R}, \alpha = \sup E$$

Suppose $\alpha \notin E$. Then

$$\begin{gathered} \forall x \in \mathbb{R}^+, \alpha - x \notin E^u \\ \Downarrow \\ \forall x \in \mathbb{R}^+, \exists p_x \in E, \alpha - x < p_x < \alpha \\ \Downarrow \\ \forall x \in \mathbb{R}^+, p_x \in \widetilde{N}_x(\alpha) \cap E \\ \Downarrow \\ \alpha \in E' \end{gathered}$$

(B) follows from **(A)**.

□

Theorem 2.4.17.

$$\forall E \subseteq |(X, d)|, E^i = E \setminus E'$$

Proof.

- (⊆) Suppose $p \in E^i$. Then

$$\begin{gathered}\exists r \in \mathbb{R}^+, E \cap N_r(p) = \{p\} \\ \Downarrow \\ p \in E \wedge E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \\ p \notin E' \\ \Downarrow \\ p \in E \setminus E'\end{gathered}$$

- (⊇) Suppose $p \in E \setminus E'$. Then

$$\begin{gathered}p \notin E' \\ \Downarrow \\ \exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(p) = \varnothing \\ \Downarrow \qquad\qquad p \in E \\ E \cap N_r(p) = \{p\} \\ \Downarrow \\ p \in E^i\end{gathered}$$

□

Lemma 2.4.18.

$$\forall E \subseteq |(X, d)|, (E^\circ)^c = \overline{E^c}$$

Proof. Suppose $p \in X$. Then

$$\begin{aligned}p \in (E^\circ)^c &\iff p \notin E^\circ \\ &\iff \forall r \in \mathbb{R}^+, N_r(p) \nsubseteq E \\ &\iff \forall r \in \mathbb{R}^+, N_r(p) \cap E^c \neq \varnothing \\ &\iff p \in \overline{E^c}\end{aligned}$$

□

Definition 2.4.19. Suppose $E \subseteq |(X, d)|$. The **boundary** ∂E of E is defined by

$$\partial E := \overline{E} \cap \overline{E^c}$$

or equivalently, by lemma 2.4.18,

$$\partial E = \overline{E} \setminus E^\circ$$

Lemma 2.4.20.

$$\forall n \in \mathbb{N}, \forall \boldsymbol{p} \in \mathbb{R}^n, \forall r \in \mathbb{R}^+, \widetilde{N}_r(\boldsymbol{p}) \neq \varnothing$$

Proof. Suppose

- $n \in \mathbb{N}$
- $\boldsymbol{p} \in \mathbb{R}^n$
- $r \in \mathbb{R}^+$

By definition 1.9.3,

$$\begin{gathered} \left\| \left(\boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1}\right) - \boldsymbol{p} \right\| = \left\| \frac{r}{2}\boldsymbol{e_1} \right\| = \frac{r}{2} < r \\ \Downarrow \\ \boldsymbol{p} + \frac{r}{2}\boldsymbol{e_1} \in \widetilde{N}_r(\boldsymbol{p}) \end{gathered}$$

where $\boldsymbol{e_1} \coloneqq (1, 0, 0, \ldots, 0) \in \mathbb{R}^n$. □

Theorem 2.4.21.

$$\forall n \in \mathbb{N}, \forall E \subseteq \mathbb{R}^n, \partial E = E^i \cup (E^c)^i \cup \left(E' \cap (E^c)'\right)$$

Proof. Suppose

- $n \in \mathbb{N}$
- $E \subseteq \mathbb{R}^n$

By definition 2.4.19,

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= \left(E \cup E'\right) \cap \left(E^c \cup (E^c)'\right) \\ &= (E \cap E^c) \cup \left(E \cap (E^c)'\right) \cup \left(E' \cap E^c\right) \cup \left(E' \cap (E^c)'\right) \end{aligned}$$

Consider

- $E = \left(E \setminus E'\right) \cup \left(E \cap E'\right)$
- $E^c = \left(E^c \setminus (E^c)'\right) \cup \left(E^c \cap (E^c)'\right)$

This implies

- $E \cap (E^c)' = \big((E \setminus E') \cap (E^c)'\big) \cup \big((E \cap E') \cap (E^c)'\big)$
- $E' \cap E^c = \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap \big(E^c \cap (E^c)'\big)\big)$

Consider

- $\big((E \cap E') \cap (E^c)'\big) \subseteq E' \cap (E^c)'$
- $\big(E' \cap \big(E^c \cap (E^c)'\big)\big) \subseteq E' \cap (E^c)'$

This implies

$$\begin{aligned}\partial E &= \varnothing \cup \big((E \setminus E') \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big) \\ &= \big((E \setminus E') \cap (E^c)'\big) \cup \big(E' \cap \big(E^c \setminus (E^c)'\big)\big) \cup \big(E' \cap (E^c)'\big)\end{aligned}$$

Suppose $\boldsymbol{p} \in E \setminus E'$. Then

$$\begin{gathered}\exists r \in \mathbb{R}^+, E \cap \widetilde{N}_r(\boldsymbol{p}) = \varnothing \\ \Downarrow \\ \widetilde{N}_r(\boldsymbol{p}) \subseteq E^c\end{gathered}$$

By lemma 2.4.20,

$$\begin{gathered}\forall s \in \mathbb{R}^+, \widetilde{N}_s(\boldsymbol{p}) \cap \widetilde{N}_r(\boldsymbol{p}) = \widetilde{N}_{\min(s,r)}(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \widetilde{N}_s(\boldsymbol{p}) \cap \left(\widetilde{N}_r(\boldsymbol{p}) \cap E^c\right) \neq \varnothing \\ \Downarrow \\ E^c \cap \widetilde{N}_s(\boldsymbol{p}) \neq \varnothing \\ \Downarrow \\ \boldsymbol{p} \in (E^c)'\end{gathered}$$

Since $\boldsymbol{p} \in E \setminus E'$ was arbitrary,

$$\begin{gathered}E \setminus E' \subseteq (E^c)' \\ \Downarrow \\ \big(E \setminus E'\big) \cap (E^c)' = E \setminus E'\end{gathered}$$

Similarly,

$$\begin{gathered}E^c \setminus (E^c)' \subseteq E' \\ \Downarrow \\ E' \cap \big(E^c \setminus (E^c)'\big) = E^c \setminus (E^c)'\end{gathered}$$

Therefore,

$$\begin{gathered}\partial E = \big(E \setminus E'\big) \cup \big(E^c \setminus (E^c)'\big) \cup \big(E' \cap (E^c)'\big) \\ \Downarrow \quad \text{Theorem 2.4.17} \\ \partial E = E^i \cup (E^c)^i \cup \big(E' \cap (E^c)'\big)\end{gathered}$$

□

Remark. Suppose $Y \subseteq |(X, d)|$. Then one can verify

$$d|_{Y \times Y} : Y \times Y \to [0, \infty)$$

is a metric on Y. Therefore, the structure

$$(Y, d|_{Y \times Y})$$

is a metric space, which is called the **subspace** of (X, d).

- (Y, d) denotes $(Y, d|_{Y \times Y})$
- $d(p, q)$ denotes $d|_{Y \times Y}(p, q)$

for the sake of simplicity.

Definition 2.4.22. Suppose $E \subseteq Y \subseteq |(X, d)|$. Then

- E is said to be **open relative to** Y if

  $$\forall p \in E, \exists r \in \mathbb{R}^+, Y \cap N_r(p) \subseteq E$$

  or equivalently,

  $$E \in \mathcal{O}(|(Y, d)|)$$

- E is said to be **closed relative to** Y if

  $$E' \cap Y \subseteq E$$

  or equivalently,

  $$E \in \mathcal{C}(|(Y, d)|)$$

Theorem 2.4.23.

$$\forall E, Y \subseteq |(X, d)|, E \in \mathcal{O}(|(Y, d)|) \iff \exists V \in \mathcal{O}(X), E = V \cap Y$$

Proof. Suppose $E, Y \subseteq X$. Then

- ($\implies$) Suppose $E \in \mathcal{O}(|(Y, d)|)$. Then

  $$\forall p \in E, \exists r_p \in \mathbb{R}^+, Y \cap N_{r_p}(p) \subseteq E$$

  Define

  $$V := \bigcup_{p \in E} N_{r_p}(p) \subseteq X$$

  By theorems 2.4.8 and 2.4.11,

  $$V \in \mathcal{O}(X)$$

  We will show $E = V \cap Y$.

- **(⊆)**

$$\begin{gathered}\forall p \in E, p \in N_{r_p}(p) \subseteq V \\ \Downarrow \\ E \subseteq V\end{gathered}$$

- **(⊇)**

$$\begin{gathered}\bigcup_{p \in E} \big(Y \cap N_{r_p}(p)\big) \subseteq E \\ \Downarrow \\ Y \cap V = Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E\end{gathered}$$

- **($\Longleftarrow$)** Suppose
  - $V \in \mathcal{O}(X)$
  - $E = V \cap Y$

  Then

$$\begin{gathered}\forall p \in E, \exists r_p \in \mathbb{R}^+, N_{r_p}(p) \subseteq V \\ \Downarrow \\ \forall p \in E, Y \cap N_{r_p}(p) \subseteq V \cap Y = E\end{gathered}$$

□

2.5 Compact Sets

Definition 2.5.1. Suppose $E \subseteq |(X, d)|$.

- A collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if
  - $\forall \alpha \in A, G_\alpha \in \mathcal{O}(X)$
  - $E \subseteq \bigcup_{\alpha \in A} G_\alpha$
- A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of $\{G_\alpha\}_{\alpha \in A}$ if
  - $S \subseteq A$
  - $E \subseteq \bigcup_{\alpha \in S} G_\alpha$

Definition 2.5.2. Suppose $E \subseteq (X, d)$. Then E is called **compact** if

$$\forall \{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X), E \subseteq \bigcup_{\alpha \in A} G_\alpha \implies \exists S \subseteq A, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} G_\alpha$$

The set of all compact sets in X is denoted by

$$\mathcal{K}(X) := \left\{ E \subseteq X \;\middle|\; \begin{array}{c} \forall \{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X), E \subseteq \bigcup_{\alpha \in A} G_\alpha \\ \implies \exists S \subseteq A, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} G_\alpha \end{array} \right\}$$

Theorem 2.5.3.

$$\forall Y \subseteq |(X, d)|, \forall E \subseteq Y, E \in \mathcal{K}(|(Y, d)|) \iff E \in \mathcal{K}(X)$$

Proof. Suppose $E \subseteq Y \subseteq X$. Then

- ($\implies$) Suppose
  - $E \in \mathcal{K}(Y)$
  - $\{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X)$
  - $E \subseteq \bigcup_{\alpha \in A} G_\alpha$

  Define
  $$\forall \alpha \in A, H_\alpha := G_\alpha \cap Y \subseteq Y$$
  By theorem 2.4.23,
  $$\forall \alpha \in A, H_\alpha \in \mathcal{O}(Y)$$
  Then
  $$E \subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha$$
  $$\Downarrow$$
  $$\exists S \subseteq A, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} H_\alpha$$
  $$\Downarrow$$
  $$E \subseteq \bigcup_{\alpha \in S} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S} G_\alpha$$

- ($\impliedby$) Suppose

- $E \in \mathcal{K}(X)$
- $\{H_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(Y)$
- $E \subseteq \bigcup_{\alpha \in A} H_\alpha$

By theorem 2.4.23,

$$\forall \alpha \in A, \exists G_\alpha \in \mathcal{O}(X), H_\alpha = G_\alpha \cap Y$$

Then

$$\begin{gathered} E \subseteq \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha \\ \Downarrow \\ \exists S \subseteq A, |S| < \aleph_0 \wedge E \subseteq \bigcup_{\alpha \in S} G_\alpha \\ \Downarrow \\ E \subseteq \left(\bigcup_{\alpha \in S} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S} (G_\alpha \cap Y) = \bigcup_{\alpha \in S} H_\alpha \end{gathered}$$

□

Theorem 2.5.4.

$$\forall E \in \mathcal{K}(|(X,d)|), E \in \mathcal{C}(X)$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $E \neq \varnothing$
- $p \in E^c$

Define

$$\forall q \in E, r_q := \frac{d(p,q)}{2} \in \mathbb{R}^+$$

Then

$$\begin{gathered} \forall q \in E, q \in N_{r_q}(q) \\ \Downarrow \\ E \subseteq \bigcup_{q \in E} N_{r_q}(q) \\ \Downarrow \quad \text{Theorem 2.4.8} \\ \exists S \subseteq E, |S| < \aleph_0 \wedge E \subseteq \bigcup_{q \in S} N_{r_q}(q) \end{gathered}$$

Define

$$T \coloneqq \{r_q \mid q \in S\} \neq \varnothing$$

Define $f : S \to T$ by

$$f(q) \coloneqq r_q$$

Then

$$\begin{gathered}
\forall t \in T, \exists q \in S, r_q = t \\
\Downarrow \\
f : S \twoheadrightarrow T \\
\Downarrow \qquad \text{Theorem 2.2.8} \\
|T| < \aleph_0 \\
\Downarrow \qquad \text{Theorem 2.2.9} \\
\exists \varepsilon \in T, \varepsilon = \min T \\
\Downarrow \\
\forall x \in N_\varepsilon(p), \forall q \in S, d(x,p) < \varepsilon \leq r_q
\end{gathered}$$

Assume

$$\exists x \in N_\varepsilon(p), \exists q \in S, x \in N_{r_q}(q)$$

Then

$$\begin{gathered}
d(p,q) \leq d(p,x) + d(x,q) < r_q + r_q = d(p,q) \\
\Downarrow \\
\perp
\end{gathered}$$

Therefore,

$$\begin{gathered}
\forall x \in N_\varepsilon(p), \forall q \in S, x \notin N_{r_q}(q) \\
\Downarrow \\
\forall q \in S, N_\varepsilon(p) \cap N_{r_q}(q) = \varnothing \\
\Downarrow \\
N_\varepsilon(p) \cap E \subseteq N_\varepsilon(p) \cap \bigcup_{q \in S} N_{r_q}(q) = \varnothing \\
\Downarrow \\
N_\varepsilon(p) \subseteq E^c
\end{gathered}$$

Since $p \in E^c$ was arbitrary,

$$\begin{gathered}
E^c \in \mathcal{O}(X) \\
\Downarrow \qquad \text{Theorem 2.4.10} \\
E \in \mathcal{C}(X)
\end{gathered}$$

$\square$

Theorem 2.5.5.

$$\forall E \in \mathcal{K}(|(X,d)|), E \in \mathcal{B}(X)$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $E \neq \varnothing$

Let $p \in E$. By theorem 1.6.2,

$$\begin{array}{cr}
\forall q \in E, \exists n \in \mathbb{N}, d(p,q) < n & \\
\Downarrow & \\
E \subseteq \bigcup_{n=1}^{\infty} N_n(p) & \\
\Downarrow & \text{Theorem 2.4.8} \\
\exists S \subseteq \mathbb{N}, |S| < \aleph_0 \wedge E \subseteq \bigcup_{n \in S} N_n(p) & \\
\Downarrow & \text{Theorem 2.2.9} \\
\exists m \in S, m = \max S & \\
\Downarrow & \\
E \subseteq N_m(p) & \\
\Downarrow & \\
E \in \mathcal{B}(X) &
\end{array}$$

□

Theorem 2.5.6.

$$\forall E \in \mathcal{K}(|(X,d)|), \forall F \in \mathcal{C}(X), F \subseteq E \implies F \in \mathcal{K}(X)$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $F \in \mathcal{C}(X)$
- $F \subseteq E$
- $\{H_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(X)$
- $F \subseteq \bigcup_{\alpha \in A} H_\alpha$

Then

$$X = F \cup F^c \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c$$

Therefore,

$$\begin{gathered} E \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c \\ \Downarrow \\ \exists B \subseteq A, |B| < \aleph_0 \land E \subseteq \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c \\ \Downarrow \\ F \subseteq \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c \\ \Downarrow \\ F \subseteq \bigcup_{\alpha \in B} H_\alpha \end{gathered}$$

Theorem 2.4.10

□

Theorem 2.5.7.

$$\forall E \in \mathcal{K}(|(X,d)|), \forall F \subseteq E, |F| \geq \aleph_0 \implies F' \cap E \neq \varnothing$$

Proof. Suppose

- $E \in \mathcal{K}(X)$
- $F \subseteq E$
- $|F| \geq \aleph_0$

Assume $F' \cap E = \varnothing$. Then

$$\begin{gathered} \forall p \in E, \exists r_p \in \mathbb{R}^+, \widetilde{N}_{r_p}(p) \cap F = \varnothing \\ \Downarrow \\ \forall p \in E, N_{r_p}(p) \cap F \subseteq \{p\} \end{gathered}$$

This implies

$$\begin{gathered}
\forall \alpha \in E, \alpha \in N_{r_\alpha}(\alpha) \\
\Downarrow \\
E \subseteq \bigcup_{\alpha \in E} N_{r_\alpha}(\alpha) \\
\Downarrow \qquad \text{Theorem 2.4.8} \\
\exists S \subseteq E, |S| < \aleph_0 \land E \subseteq \bigcup_{\alpha \in S} N_{r_\alpha}(\alpha) \\
\Downarrow \\
F \subseteq \bigcup_{\alpha \in S} N_{r_\alpha}(\alpha) \\
\Downarrow \\
F = \left(\bigcup_{\alpha \in S} N_{r_\alpha}(\alpha)\right) \cap F = \bigcup_{\alpha \in S} (N_{r_\alpha}(\alpha) \cap F) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S \\
\Downarrow \qquad \text{Theorem 2.2.5} \\
|F| < \aleph_0 \\
\Downarrow \\
\perp
\end{gathered}$$

Therefore,

$$F' \cap E \neq \varnothing$$

□

Theorem 2.5.8.

$$\forall \{K_\alpha\}_{\alpha \in A} \subseteq \mathcal{K}(|(X,d)|),$$
$$\left(\forall B \subseteq A, |B| < \aleph_0 \implies \bigcap_{\alpha \in B} K_\alpha \neq \varnothing\right) \implies \bigcap_{\alpha \in A} K_\alpha \neq \varnothing$$

Proof. Suppose

- $\{K_\alpha\}_{\alpha \in A} \subseteq \mathcal{K}(X)$
- $\forall B \subseteq A, |B| < \aleph_0 \implies \bigcap_{\alpha \in B} K_\alpha \neq \varnothing$
- $A \neq \varnothing$

By theorem 2.5.4,

$$\begin{gathered}\forall \alpha \in A, K_\alpha \in \mathcal{C}(X) \\ \Downarrow \\ \forall \alpha \in A, (K_\alpha)^c \in \mathcal{O}(X)\end{gathered} \qquad \text{Theorem 2.4.10}$$

Assume $\bigcap_{\alpha\in A} K_\alpha = \varnothing$. Then

$$X = \left(\bigcap_{\alpha\in A} K_\alpha\right)^c = \bigcup_{\alpha\in A} (K_\alpha)^c$$

Let $\beta \in A$. Then

$$\begin{gathered}K_\beta \subseteq \bigcup_{\alpha\in A} (K_\alpha)^c \\ \Downarrow \\ \exists B \subseteq A, |B| < \aleph_0 \wedge K_\beta \subseteq \bigcup_{\alpha\in B} (K_\alpha)^c \\ \Downarrow \\ K_\beta \subseteq \left(\bigcap_{\alpha\in B} K_\alpha\right)^c \\ \Downarrow \\ K_\beta \cap \left(\bigcap_{\alpha\in B} K_\alpha\right) = \varnothing \\ \Downarrow \\ \bigcap_{\alpha\in B\cup\{\beta\}} K_\alpha = \varnothing\end{gathered}$$

By theorem 2.2.16,

$$\begin{gathered}|B \cup \{\beta\}| < \aleph_0 \\ \Downarrow \\ \bot\end{gathered}$$

Therefore,

$$\bigcap_{\alpha\in A} K_\alpha \neq \varnothing$$

□

Lemma 2.5.9.

$$(\forall n \in \mathbb{N}, -\infty < a_n \leq a_{n+1} \leq b_{n+1} \leq b_n < +\infty) \implies \bigcap_{n=1}^{\infty} [a_n, b_n] \neq \varnothing$$

Proof. Suppose

$$\forall n \in \mathbb{N}, -\infty < a_n \leq a_{n+1} \leq b_{n+1} \leq b_n < +\infty$$

Then

- $\forall n, m \in \mathbb{N}, n \leq m \implies a_n \leq a_m \leq b_m$
- $\forall n, m \in \mathbb{N}, n > m \implies a_n \leq b_n \leq b_m$

Therefore,

$$\begin{gathered}\forall m \in \mathbb{N}, b_m \in (\{a_n\}_{n=1}^{\infty})^u \\ \Downarrow \qquad \text{Axiom 1.6.1} \\ \exists \alpha \in \mathbb{R}, \alpha = \sup\{a_n\}_{n=1}^{\infty} \\ \Downarrow \\ \forall m \in \mathbb{N}, a_m \leq \alpha \leq b_m \\ \Downarrow \\ \alpha \in \bigcap_{n=1}^{\infty}[a_n, b_n]\end{gathered}$$

□

Theorem 2.5.10.

$$\forall k \in \mathbb{N}, (\forall n \in \mathbb{N}, \forall j \in [1,k]_{\mathbb{N}}, -\infty < a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j} < +\infty)$$
$$\implies \bigcap_{n=1}^{\infty}\left(\prod_{j=1}^{k}[a_{n,j}, b_{n,j}]\right) \neq \varnothing$$

Proof. Suppose

- $k \in \mathbb{N}$
- $\forall n \in \mathbb{N}, \forall j \in [1,k]_{\mathbb{N}}, -\infty < a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j} < +\infty$

By lemma 2.5.9,

$$\begin{gathered}\forall j \in [1,k]_{\mathbb{N}}, \bigcap_{n=1}^{\infty}[a_{n,j}, b_{n,j}] \neq \varnothing \\ \Downarrow \\ \forall j \in [1,k]_{\mathbb{N}}, \exists \alpha_j \in \bigcap_{n=1}^{\infty}[a_{n,j}, b_{n,j}] \\ \Downarrow \\ (\alpha_1, \alpha_2, \ldots, \alpha_k) \in \bigcap_{n=1}^{\infty}\left(\prod_{j=1}^{k}[a_{n,j}, b_{n,j}]\right)\end{gathered}$$

□

Theorem 2.5.11.

$$\forall k \in \mathbb{N}, (\forall j \in [1,k]_{\mathbb{N}}, -\infty < a_j \leq b_j < +\infty) \\ \implies \prod_{j=1}^{k} [a_j, b_j] \in \mathcal{K}(\mathbb{R}^k)$$

Proof. Suppose

- $k \in \mathbb{N}$
- $\forall j \in [1,k]_{\mathbb{N}}, -\infty < a_j \leq b_j < +\infty$

Define

- $\forall j \in [1,k]_{\mathbb{N}}, (a_{1,j}, b_{1,j}) \coloneqq (a_j, b_j)$
- $I_1 \coloneqq \displaystyle\prod_{j=1}^{k} [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k$
- $\forall j \in [1,k]_{\mathbb{N}}, \forall m \in \{0,1\}, I_{1,j,m} \coloneqq \begin{cases} \left[a_{1,j}, \dfrac{a_{1,j}+b_{1,j}}{2}\right], & \text{if } m = 0 \\ \left[\dfrac{a_{1,j}+b_{1,j}}{2}, b_{1,j}\right], & \text{otherwise} \end{cases}$
- $S \coloneqq \displaystyle\prod_{j=1}^{k} \{0,1\} = \left\{ \boldsymbol{m} \in \mathbb{R}^k \;\middle|\; \forall j \in [1,k]_{\mathbb{N}}, m_j \in \{0,1\} \right\}$

Then

$$\forall \boldsymbol{m} \in S, \forall j \in [1,k]_{\mathbb{N}}, I_{1,j,m_j} \subseteq [a_{1,j}, b_{1,j}] \\ \Downarrow \\ \forall \boldsymbol{m} \in S, \prod_{j=1}^{k} I_{1,j,m_j} \subseteq I_1$$

Suppose

- $\{G_\alpha\}_{\alpha \in A} \subseteq \mathcal{O}(\mathbb{R}^k)$
- $I_1 \subseteq \displaystyle\bigcup_{\alpha \in A} G_\alpha$

Assume

- $\nexists B \subseteq A, |B| < \aleph_0 \wedge I_1 \subseteq \displaystyle\bigcup_{\alpha \in B} G_\alpha$

- $\forall \boldsymbol{m} \in S, \exists A_{1,\boldsymbol{m}} \subseteq A, |A_{1,\boldsymbol{m}}| < \aleph_0 \wedge \prod_{j=1}^{k} I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{1,\boldsymbol{m}}} G_\alpha$

Define

$$B_1 := \bigcup_{\boldsymbol{m} \in S} A_{1,\boldsymbol{m}}$$

By theorem 2.2.19,

$$\begin{gathered} |S| < \aleph_0 \\ \Downarrow \\ |B_1| < \aleph_0 \end{gathered} \qquad \text{Theorem 2.2.16}$$

Then

$$\begin{gathered} I_1 = \bigcup_{\boldsymbol{m} \in S} \left(\prod_{j=1}^{k} I_{1,j,m_j} \right) \\ \Downarrow \\ I_1 \subseteq \bigcup_{\boldsymbol{m} \in S} \left(\bigcup_{\alpha \in A_{1,\boldsymbol{m}}} G_\alpha \right) = \bigcup_{\alpha \in B_1} G_\alpha \\ \Downarrow \\ \perp \end{gathered}$$

Therefore,

$$\exists \boldsymbol{m_1} \in S, \nexists A_{1,\boldsymbol{m_1}} \subseteq A, |A_{1,\boldsymbol{m_1}}| < \aleph_0 \wedge \prod_{j=1}^{k} I_{1,j,(m_1)_j} \subseteq \bigcup_{\alpha \in A_{1,\boldsymbol{m_1}}} G_\alpha$$

Define

- $\forall j \in [1,k]_{\mathbb{N}}, a_{2,j} := \begin{cases} a_{1,j}, & \text{if } (m_1)_j = 0 \\ \dfrac{a_{1,j} + b_{1,j}}{2}, & \text{otherwise} \end{cases}$
- $\forall j \in [1,k]_{\mathbb{N}}, b_{2,j} := \begin{cases} \dfrac{a_{1,j} + b_{1,j}}{2}, & \text{if } (m_1)_j = 0 \\ b_{1,j}, & \text{otherwise} \end{cases}$
- $I_2 := \prod_{j=1}^{k} [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k$
- $I_{2,j,m} := \begin{cases} \left[a_{2,j}, \dfrac{a_{2,j} + b_{2,j}}{2} \right], & \text{if } m = 0 \\ \left[\dfrac{a_{2,j} + b_{2,j}}{2}, b_{2,j} \right], & \text{otherwise} \end{cases}$

Then

$$\begin{gathered}\forall j \in [1,k]_{\mathbb{N}}, [a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \\ \Downarrow \\ I_2 \subseteq I_1\end{gathered}$$

Assume

$$\forall \boldsymbol{m} \in S, \exists A_{2,\boldsymbol{m}} \subseteq A, |A_{2,\boldsymbol{m}}| < \aleph_0 \wedge \prod_{j=1}^{k} I_{2,j,m_j} \subseteq \bigcup_{\alpha \in A_{2,\boldsymbol{m}}} G_\alpha$$

Define

$$B_2 := \bigcup_{\boldsymbol{m} \in S} A_{2,\boldsymbol{m}}$$

By theorem 2.2.19,

$$\begin{gathered}|S| < \aleph_0 \\ \Downarrow \\ |B_2| < \aleph_0\end{gathered} \qquad \text{Theorem 2.2.16}$$

Then

$$\begin{gathered}I_2 = \bigcup_{\boldsymbol{m} \in S} \left(\prod_{j=1}^{k} I_{2,j,m_j} \right) \\ \Downarrow \\ I_2 \subseteq \bigcup_{\boldsymbol{m} \in S} \left(\bigcup_{\alpha \in A_{2,\boldsymbol{m}}} G_\alpha \right) = \bigcup_{\alpha \in B_2} G_\alpha \\ \Downarrow \\ \perp\end{gathered}$$

Therefore,

$$\exists \boldsymbol{m_2} \in S, \nexists A_{\boldsymbol{m_2}} \subseteq A, |A_{\boldsymbol{m_2}}| < \aleph_0 \wedge \prod_{j=1}^{k} I_{2,j,(m_2)_j} \subseteq \bigcup_{\alpha \in A_{\boldsymbol{m_2}}} G_\alpha$$

By repeating this process,

- $\forall n \in \mathbb{N}, \forall j \in [1,k]_{\mathbb{N}}, a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$
- $\forall n \in \mathbb{N}, \forall j \in [1,k]_{\mathbb{N}}, b_{n+1,j} - a_{n+1,j} = \dfrac{1}{2}(b_{n,j} - a_{n,j})$
- $\forall n \in \mathbb{N}, \nexists A_n \subseteq A, |A_n| < \aleph_0 \wedge I_n \subseteq \displaystyle\bigcup_{\alpha \in A_n} G_\alpha$

Define

$$\delta := \sqrt{\sum_{j=1}^{k} (b_{1,j} - a_{1,j})^2} \in [0, +\infty)$$

Suppose $\delta \neq 0$. Then

$$\forall n \in \mathbb{N}, \forall \boldsymbol{x}, \boldsymbol{y} \in I_n, \|\boldsymbol{x} - \boldsymbol{y}\| \leq \sqrt{\sum_{j=1}^{k} (b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}}$$

By theorem 2.5.10,

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing$$

Let $\boldsymbol{\beta} \in \bigcap_{n=1}^{\infty} I_n$. Then

$$\begin{array}{cr}
\boldsymbol{\beta} \in \bigcup_{\alpha \in A} G_\alpha & \\
\Downarrow & \\
\exists \alpha_0 \in A, \boldsymbol{\beta} \in G_{\alpha_0} & \\
\Downarrow & \\
\exists \varepsilon \in \mathbb{R}^+, N_\varepsilon(\boldsymbol{\beta}) \subseteq G_{\alpha_0} & \\
\Downarrow & \text{Theorem 1.6.4} \\
\exists N \in \mathbb{N}, \dfrac{1}{2^N} < \dfrac{\varepsilon}{2\delta} & \\
\Downarrow & \boldsymbol{\beta} \in I_N \\
\forall \boldsymbol{\gamma} \in I_N, \|\boldsymbol{\beta} - \boldsymbol{\gamma}\| \leq \dfrac{\delta}{2^{N-1}} < \varepsilon & \\
\Downarrow & \\
I_N \subseteq N_\varepsilon(\boldsymbol{\beta}) & \\
\Downarrow & \\
I_N \subseteq \bigcup_{\alpha \in \{\alpha_0\}} G_\alpha & \\
\Downarrow & \\
\perp &
\end{array}$$

Therefore,

$$\exists B \subseteq A, |B| < \aleph_0 \wedge I_1 \subseteq \bigcup_{\alpha \in B} G_\alpha$$

$\square$

Corollary 2.5.12.

$$\forall a, b \in \mathbb{R}, a \leq b \implies [a, b] \in \mathcal{K}(\mathbb{R})$$

Theorem 2.5.13.

$$\forall n \in \mathbb{N}, \forall E \subseteq \mathbb{R}^n, E \in \mathcal{K}(\mathbb{R}^n) \iff E \in \mathcal{C}(\mathbb{R}^n) \wedge E \in \mathcal{B}(\mathbb{R}^n)$$

*This theorem is called the **Heine-Borel Theorem**.*

Proof. Suppose $n \in \mathbb{N}$.

- ($\implies$) The proof follows directly from theorems 2.5.4 and 2.5.5.
- ($\impliedby$) Suppose
  - $E \in \mathcal{C}(\mathbb{R}^n)$
  - $E \in \mathcal{B}(\mathbb{R}^n)$

  Then
  $$\exists r \in \mathbb{R}^+, \exists \boldsymbol{p} \in \mathbb{R}^n, E \subseteq N_r(\boldsymbol{p})$$
  Define
  $$I := \prod_{j=1}^{n} [p_j - r, p_j + r] \subseteq \mathbb{R}^n$$
  By theorem 2.5.11,
  $$\begin{array}{c} I \in \mathcal{K}(\mathbb{R}^n) \\ \Downarrow \\ E \in \mathcal{K}(\mathbb{R}^n) \end{array} \qquad \text{Theorem 2.5.6}$$

$\square$

2.6 Connected Sets

Definition 2.6.1. Suppose (X, d) is a metric space and $E \subseteq X$. E is called **connected** if there is no pair of sets $A, B \subseteq X$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Such pair of sets A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.6.2. *Suppose $E \subseteq \mathbb{R}$. Then E is connected if and only if*

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Proof. ($\Longrightarrow$) Suppose $E \subseteq \mathbb{R}$ is connected. Assume there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Define

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \varnothing$ and $B \neq \varnothing$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\Longleftarrow$) Suppose $E \subseteq \mathbb{R}$ satisfies

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume E is disconnected. Then there exist separated sets $A, B \subseteq \mathbb{R}$ such that

- $\varnothing \neq A \subseteq E$ and $\varnothing \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \varnothing$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$. Without loss of generality, suppose that $a < b$. Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \varnothing$.

- $A \cap [a, b]$ is bounded above by b.

Then the *existence* of α is guaranteed by the *least upper bound property*. By theorem 2.4.16, we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By definition 2.4.4, it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \overline{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$.
- $\alpha \in \overline{A} \implies \alpha \notin B \implies \alpha \neq b$.

If $\alpha \notin A$, then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$ and $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \overline{B}.$$

Then there exists $r \in (0, b - \alpha)$ such that

$$\widetilde{N}_r(\alpha) \cap B = \varnothing \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that E is connected. $\square$

Chapter 3

Sequences and Series

3.1 Convergent Sequences

Definition 3.1.1. Suppose (X, d) is a metric space. A **sequence** $(x_n)_{n=1}^{\infty}$ is a function $f : \mathbb{N} \to X$, where $x_n \coloneqq f(n)$ for each $n \in \mathbb{N}$.

A sequence $(x_n)_{n=1}^{\infty}$ is said to be **convergent** if there exists $x \in X$ such that

$$\forall \varepsilon \in \mathbb{R}^+, \exists N \in \mathbb{N}, \forall n \geq N, d(x_n, x) < \varepsilon$$

In this case, we say that

- $(x_n)_{n=1}^{\infty}$ **converges** to x.
- x is the **limit** of $(x_n)_{n=1}^{\infty}$.

We denote the limit of $(x_n)_{n=1}^{\infty}$ by

- $\lim\limits_{n\to\infty} x_n = x$.
- $x_n \to x$ as $n \to \infty$.

If $(x_n)_{n=1}^{\infty}$ is not convergent, it is said to be **divergent**.

Example 3.1.2. In $(\mathbb{R}, \|\cdot\|)$, define $(x_n)_{n=1}^{\infty}$ by

$$x_n \coloneqq \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each $\varepsilon \in \mathbb{R}^+$, we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left|\frac{1}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

Therefore, $(x_n)_{n=1}^{\infty}$ converges to 0.

In the subspace $\left(\mathbb{R}^+, \|\cdot\|\right)$ of $(\mathbb{R}, \|\cdot\|)$, define $(x_n)_{n=1}^{\infty}$ by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By theorem 1.6.2, for each $y \in \mathbb{R}^+$, there exists $N_y \in \mathbb{N}$ such that $N_y > \dfrac{1}{y}$. Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left|\frac{1}{n} - y\right| > \left|\frac{1}{N_y} - y\right| = \varepsilon_y.$$

Therefore, $(x_n)_{n=1}^{\infty}$ diverges in $\left(\mathbb{R}^+, \|\cdot\|\right)$.

Remark. Suppose (X, d) is a metric space. In this textbook, $(x_n)_{n=1}^{\infty}$ represents a sequence in (X, d), while $\{x_n\}_{n=1}^{\infty}$ represents a set defined by

$$\{x_n\}_{n=1}^{\infty} := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

Definition 3.1.3. Suppose (X, d) is a metric space. A sequence $(x_n)_{n=1}^{\infty}$ is said to be **bounded** if $\{x_n\}_{n=1}^{\infty}$ is bounded.

Theorem 3.1.4. *Suppose (X, d) is a metric space. Suppose $(x_n)_{n=1}^{\infty}$ is a convergent sequence. Then we have*

- ***(A)*** *$(x_n)_{n=1}^{\infty}$ is bounded.*
- ***(B)*** *The limit of $(x_n)_{n=1}^{\infty}$ is unique.*

Proof. Let x be a limit of $(x_n)_{n=1}^{\infty}$.

- **(A)** Let $\varepsilon = 1$. Then there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, x) < 1.$$

Define

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let $f : [1, N]_{\mathbb{N}} \to S$ defined by

$$f(k) = d(x_k, x) \quad \text{for all } k \in [1, N]_{\mathbb{N}}.$$

Then f is surjective. By lemma 2.2.7, S is finite. Then, by Theorem 2.2.9, S has a greatest element, say $M := \max S$. Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $(x_n)_{n=1}^{\infty}$ is bounded.

(B) Suppose y be a limit of $(x_n)_{n=1}^{\infty}$. Assume $x \neq y$. Then we have $r := d(x, y) > 0$. Let $\varepsilon = \frac{r}{2}$. Then we have

- There exists $N_x \in \mathbb{N}$ such that $n \geq N_x \implies d(x_n, x) < \frac{r}{2}$.
- There exists $N_y \in \mathbb{N}$ such that $n \geq N_y \implies d(x_n, y) < \frac{r}{2}$.

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x, y) \leq d(x, x_n) + d(x_n, y) < r,$$

which is a contradiction. Therefore, we have $x = y$.

$\square$

Theorem 3.1.5. *Suppose (X, d) is a metric space. Then the following statements hold:*

(A) *A sequence $(x_n)_{n=1}^{\infty}$ converges to $x \in X$ if and only if*

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) *Suppose $E \subseteq X$. If x is a limit point of E, then there exists $(x_n)_{n=1}^{\infty}$ in (X, d) such that*

$$\lim_{n \to \infty} x_n = x.$$

Proof.

(A) **($\implies$)** Suppose $(x_n)_{n=1}^{\infty}$ converge to $x \in X$. Then, for each $\varepsilon \in \mathbb{R}^+$, there exists $M_\varepsilon \in \mathbb{N}$ such that

$$n \geq M_\varepsilon \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_\varepsilon}^{\infty} \subseteq N_\varepsilon(x).$$

($\Longleftarrow$) Suppose $(x_n)_{n=1}^{\infty}$ in (X, d) such that

For every $\varepsilon \in \mathbb{R}^+$, $N_\varepsilon(x)$ contains all but finitely many terms of $(x_n)_{n=1}^{\infty}$.

(B) Suppose $E \subseteq X$ and x is a limit point of E. Then for every $\varepsilon \in \mathbb{R}^+$, we have $(N_\varepsilon(x) - \{x\}) \cap E \neq \varnothing$. Thus we can choose $(y_n)_{n=1}^{\infty}$ in (X, d) such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that $\lim_{n \to \infty} y_n = x$.

$\square$

3.2 Subsequences

3.3 Cauchy Sequences

3.4 Upper and Lower Limits

3.5 Some Special Sequences

3.6 Series

3.7 Series of Nonnegative Terms

3.8 The Number e

3.9 The Root and Ratio Tests

3.10 Power Series

3.11 Summation by Parts

3.12 Absolute Convergence

3.13 Addition and Multiplication of Series

3.14 Rearrangements

Chapter 4

Continuity

4.1 Limits of Functions

4.2 Continuous Functions

4.3 Continuity and Compactness

4.4 Continuity and Connectedness

4.5 Discontinuities

4.6 Monotonic Functions

4.7 Infinite Limits and Limits at Infinity

Chapter 5

Differentiation

5.1 The Derivative of a Real Function

5.2 Mean Value Theorems

5.3 The Continuity of Derivatives

5.4 L'Hospital's Rule

5.5 Derivatives of Higher Order

5.6 Taylor's Theorem

5.7 Differentiation of Vector-valued Functions

Chapter 6

The Riemann-Stieltjes Integral

6.1 Definition and Existence of the Integral

6.2 Properties of the Integral

6.3 Integration and Differentiation

6.4 Integration of Vector-valued Functions

6.5 Rectifiable Curves

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